Social Networks and their Analysis

REAL-WORLD COMPLEX NETWORKS, COMMUNITY DETECTION AND LINK PREDICTION

BY: IVETA MRAZOVA

Social Networks and their Analysis

Link Analysis

- *A Brief Review of Actor-Level and Network-Level Measures*
- *Hubs and Authorities, the HITS Algorithm*
- *The PageRank Algorithm*
- *PageRank – Revisited*
- *HITS – Revisited*
- *Themed Communities and Their Discovery*

Real-World Complex Networks, Community Detection, and Link Prediction

- Real-World Complex Networks and Their Properties
- Community Detection in Social Networks
- Link Prediction

Social Networks and their Analysis

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- **Real-World Complex Networks and Their Properties**
- *Community Detection in Social Networks*
  - *Communities and Strategies for Their Detection*
  - *Minimum Cut Methods*
  - *The Girvan-Newman Algorithm*
  - *The Louvain Algorithm*
  - *Other Community Detection Techniques*
    - *Node-Centric and Group-Centric Approaches*
    - *Network-Centric Community Detection*
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  - *Neighborhood-Based Link Prediction Measures*
  - *The Katz Measure*
  - *Link Prediction as a Classification Problem*
  - *Summary*

Properties of Real-World Complex Networks

~ non-random and non-regular graphs with unique features

characterized by:

1. **Small-world effect**
2. **Transitivity or clustering**
3. **Power-law degree distributions**
4. **Network resilience**
5. **Mixing patterns**
6. **Community structure**

Examples of such networks:

- Social networks
- Information or knowledge networks
- Technological networks
- Biological networks

Properties of Real-World Complex Networks

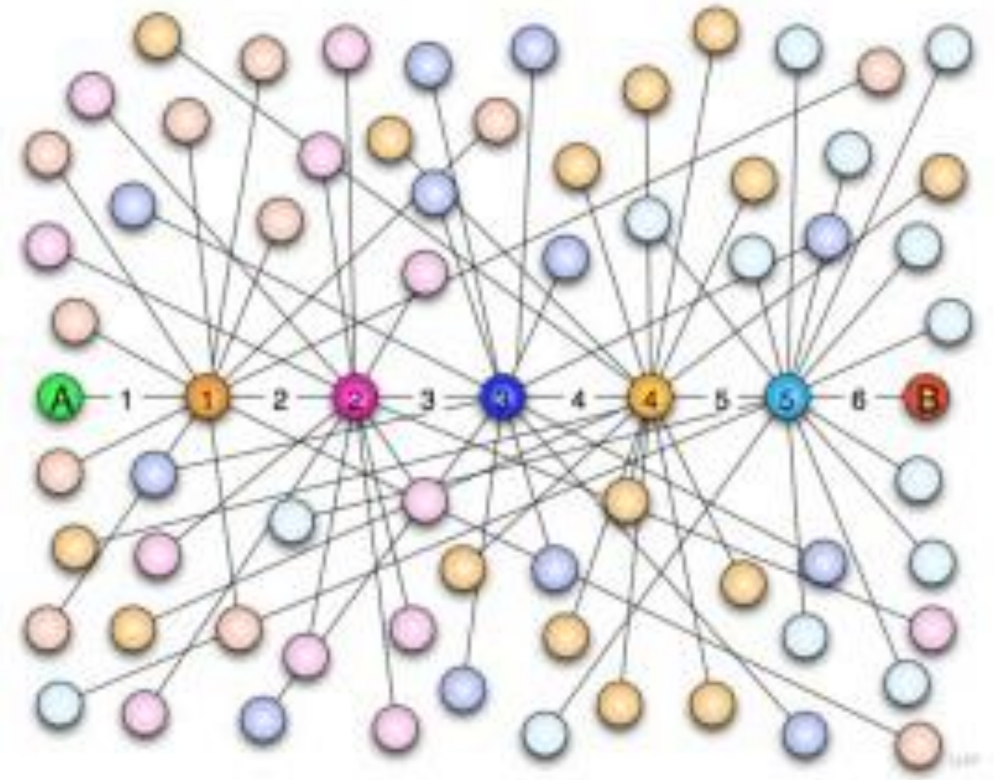
1. Small-world effect:

- Stanley Milgram (American social psychologist) was the first to point out the existence of *small-world effects* in real social networks (*Milgram experiment,* 1960s)
- Milgram asked random participants (about 300) to pass a letter to someone they knew on a first-name basis in an attempt to get it to an assigned target person.
- **The goal of the experiment:** test the speculative idea that a pair of apparently distant individuals in most networks are connected by a small number of acquaintances
- Through this experiment, Milgram was able to show that the median path length of the paths that succeeded in reaching the target person was 6

Properties of Real-World Complex Networks

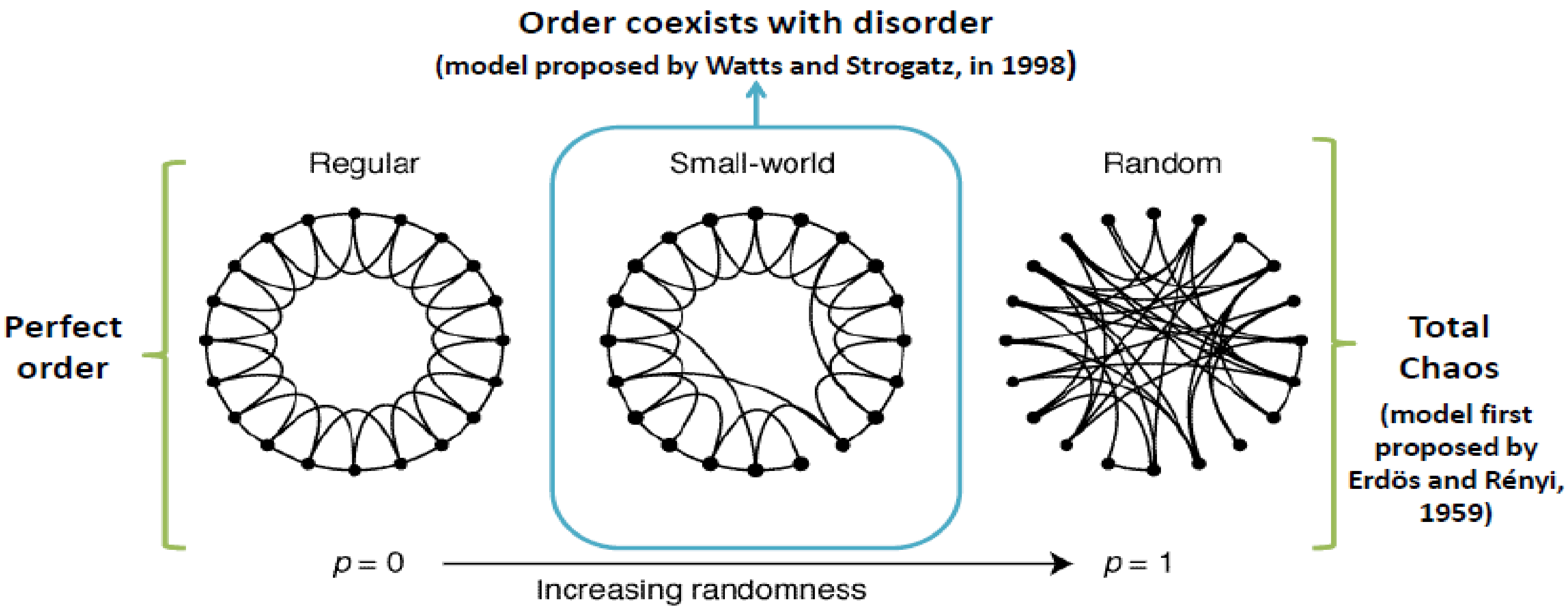
1. Small-world effect:

- based on the *six degrees of separation* concept, which states that everyone is, on average, six steps away from any other person in the World
- in a broad sense, Milgram´s conclusion indicates that social networks tend to be characterized by very short paths between randomly chosen pairs of people.
- important consequences, e.g., for the speed at which information and diseases spread



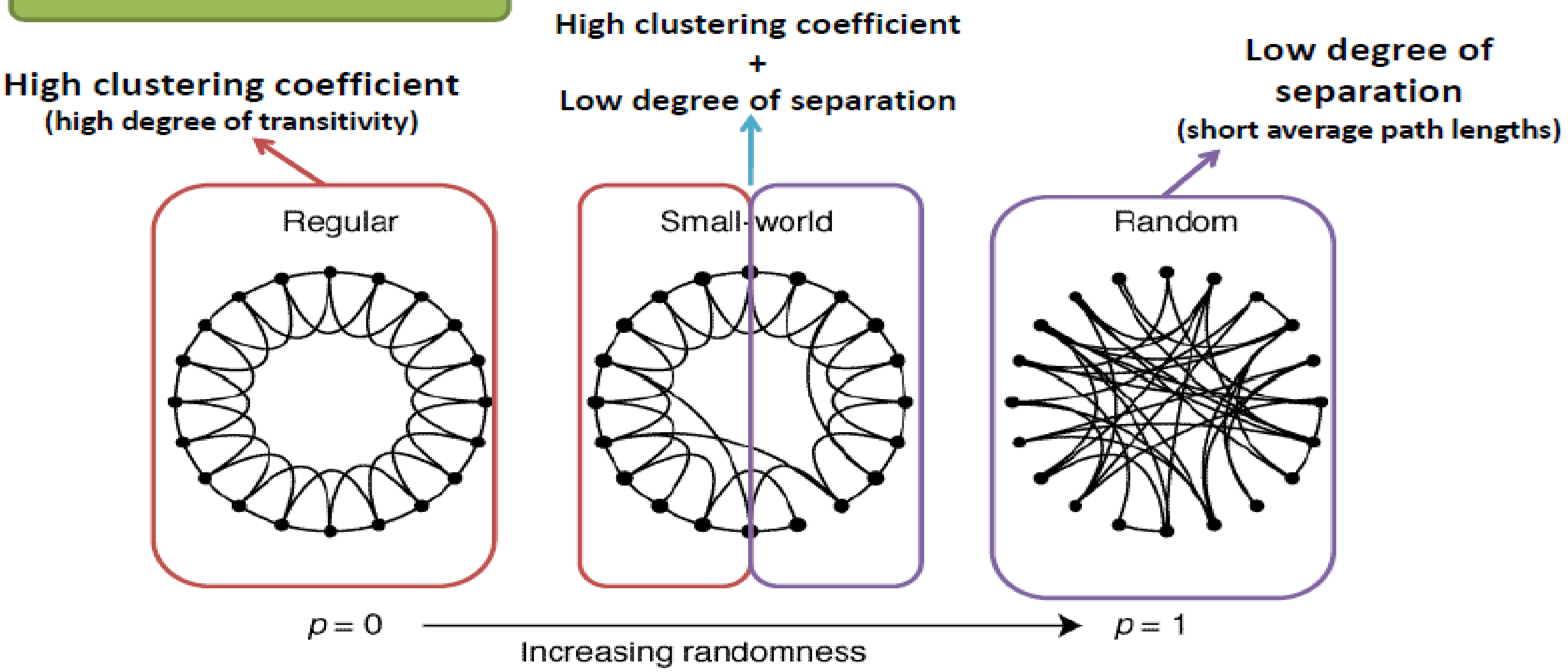
Properties of Real-World Complex Networks

1. Small-world effect:



Properties of Real-World Complex Networks

1. Small-world effect



Properties of Real-World Complex Networks

2. Transitivity or Clustering:

- In real-world networks, especially in social ones, there is a high probability of finding complete sub-networks, within larger networks, where all nodes are connected to each other (everyone knows everyone)

3. Power-law degree distributions:

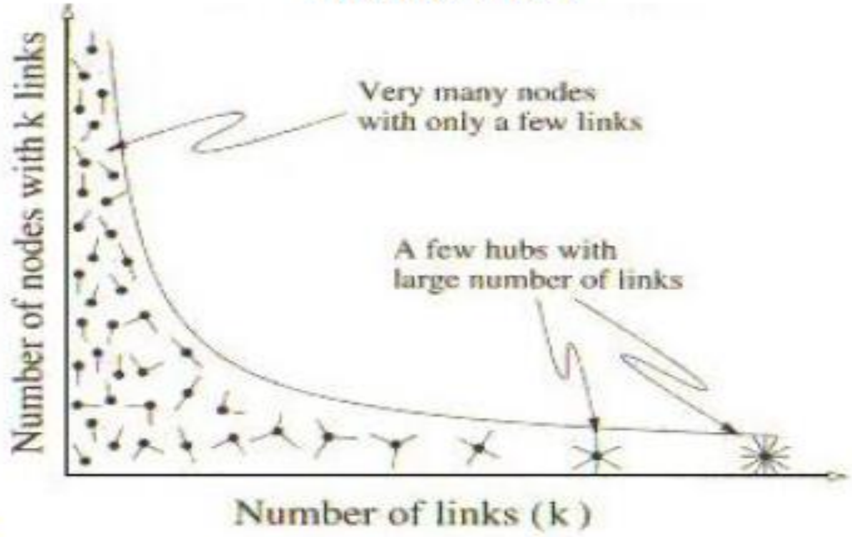
- **Degree distribution:** probability distribution of the degrees of nodes over the whole network (~ a histogram that shows the fraction of nodes in the network that have degree k, $k = 1, 2, 3, \dots, k_{max}$)
- real networks have degree distributions that are quite different from other networks, such as random graphs

Properties of Real-World Complex Networks

3. Power-law degree distributions

Random graphs	Real-world networks
↓	↓
Homogeneous degree distribution	Heterogeneous degree distribution
↓	↓
Binomial degree distribution (Poisson in the limit of graph size)	**Power-law** degree distribution (at least asymptotically)
	↓
	Scale-free networks

Power-law degree distribution



Properties of Real-World Complex Networks

3. Power-law degree distribution:

Scale-free networks (Barabasi and Albert, 1999):

~ networks whose degree distribution follows a power-law (e.g., World Wide Web, citation networks, biological networks, some social networks, etc.)

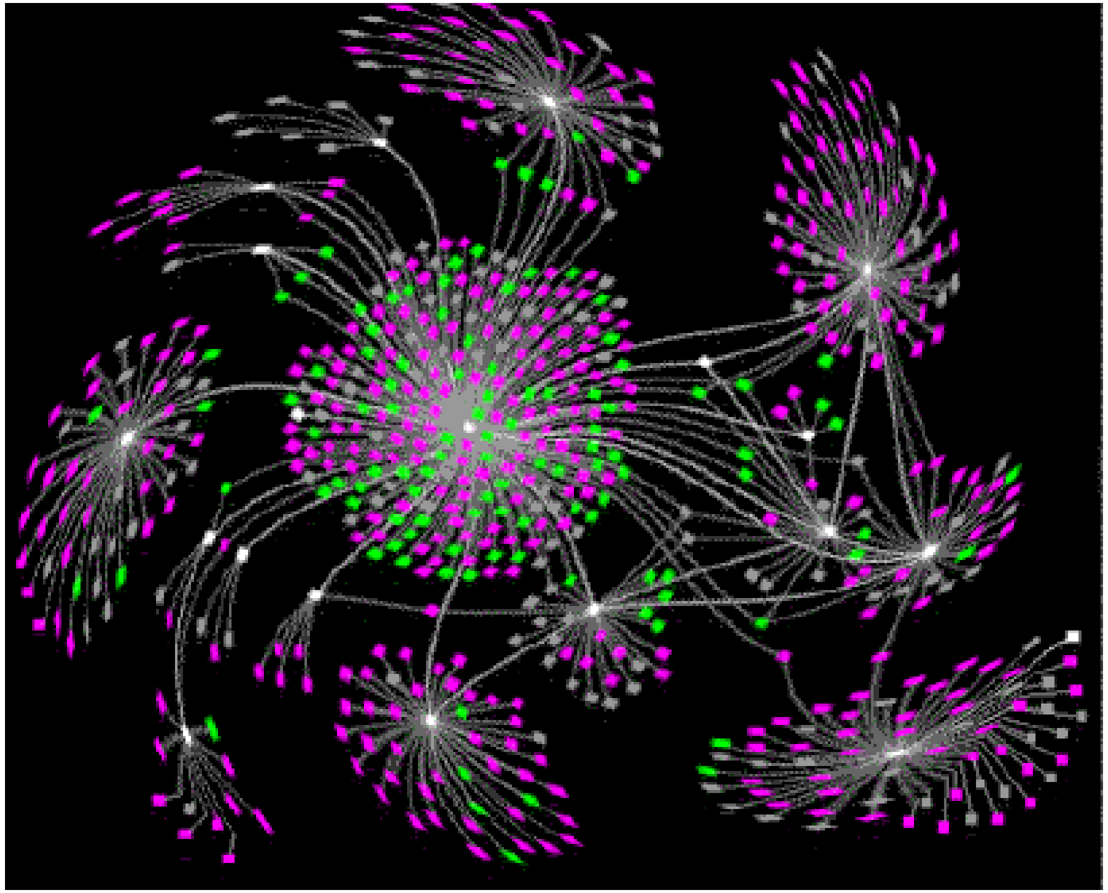
- Power-law distributions usually arise when the amount you get of something depends on the amount you already have; in topological terms this reflects in a network with few highly-connected nodes and a large number of nodes with low degree
- The mechanism behind this kind of degree distribution was referred to as **the rich-get-richer and the poor-get-poorer** strategy or, equivalently, the **Matthew´s effect**; Prince called it **cumulative advantage**, and, more recently, Barabasi and Albert used the expression **preferential attachment**

Properties of Real-World Complex Networks

3. Power-law degree distribution:

Scale-free networks (Barabasi and Albert, 1999):

- The mechanism of **preferential attachment** is easily explained by the fact that **new nodes** (e.g., individuals) entering the network **tend to connect to well-connected nodes**, which are often associated with central and prestigious positions (e.g., individuals with more status, popularity, knowledge, money, etc.) in the network
- Such highly-connected nodes are called ***hubs***.



Properties of Real-World Complex Networks

4. Network Resilience:

~ measures the impact on the connectivity of the network when one or more nodes are removed

- Most networks are robust against random vertex removal, but considerably less robust to targeted removal of the highest-degree vertices
- Also, when gatekeepers are deleted, there occur significant changes in the network´s communication ability between pairs of nodes, since some nodes become disconnected.
- In real-world networks, the removal of a single node is not cause for alarm, since it rarely affects the original network structure; in such cases, it is more appropriate to test the network´s resilience by removing a certain percentage of nodes.

Properties of Real-World Complex Networks

5. Mixing Patterns:

- In some networks, where different types of nodes coexist, it is common to observe a certain selectivity in the establishment of connections
- **Selective linking** is called **assortative mixing** (or *homophily*), and a classic example is mixing by race
- Real networks show higher tendencies for assortative mixing

Properties of Real-World Complex Networks

6. Community Structure:

- Most real social networks show **community structure**
- community structure usually arises as a consequence of both global and local heterogeneity of edges´ distribution in a graph

➔ we often find high concentrations of edges within some areas of the graph, which we call **communities**, and low concentrations of edges between those regions

Preferential Attachment

~ in some networks, new edges tend to grow towards nodes with an already high degree in time

➔ the degree of some nodes thus increases disproportionately compared to the other ones

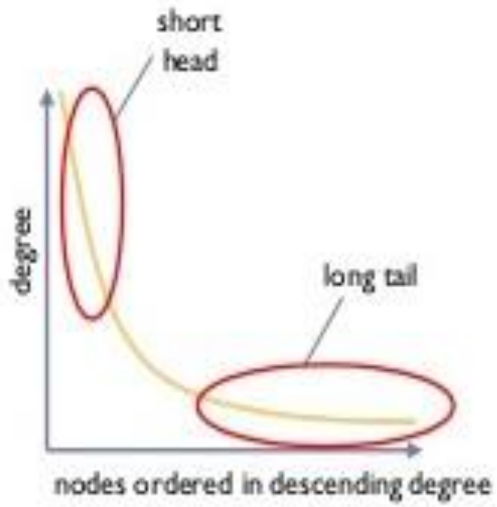
The resulting network has few highly connect-ed nodes and many nodes with a low degree

- **long-tailed degree distribution**
- **small-world structure**

➔ *transitivity and strong/weak tie charact-eristics are common and can lead to* small-world *structures*

× ***but they are not necessary to explain it***

Example of a network with preferential attachment



Sketch of long-tailed degree distribution

Reasons for Preferential Attachment

- **Popularity ~ *'the rich get richer'***
  - we want to be associated with popular people, ideas, and items,
  - this further increases their popularity, irrespective of any objective, measurable characteristics
- **Quality ~ *'the good get better'***
  - we evaluate people and everything else based on objective quality criteria, so higher quality nodes will naturally attract more attention faster
- **Mixed model**

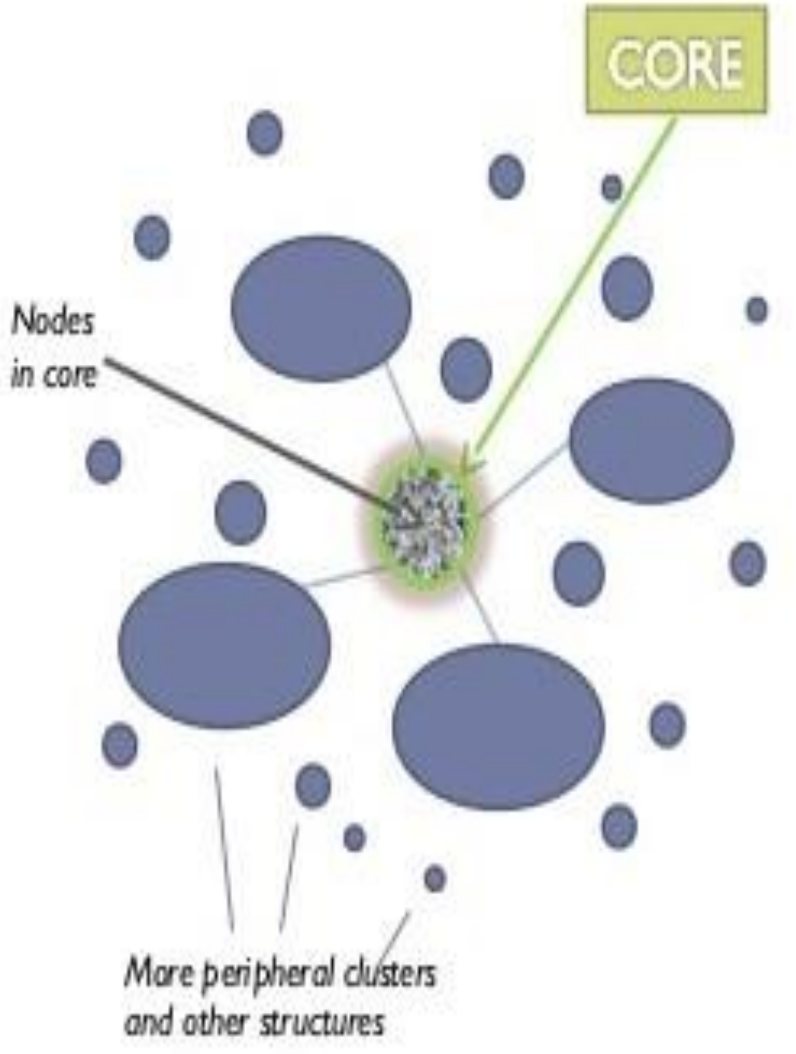
  ~ *´It is impossible to predict who will become a star, even if quality matters´*
  - among nodes of similar attributes, those that reach critical mass first will become 'stars' with many friends and followers ('halo effect')

Core-Periphery Structures

Centralization measure

~ measures the degree to which a social network is centralized or decentralized

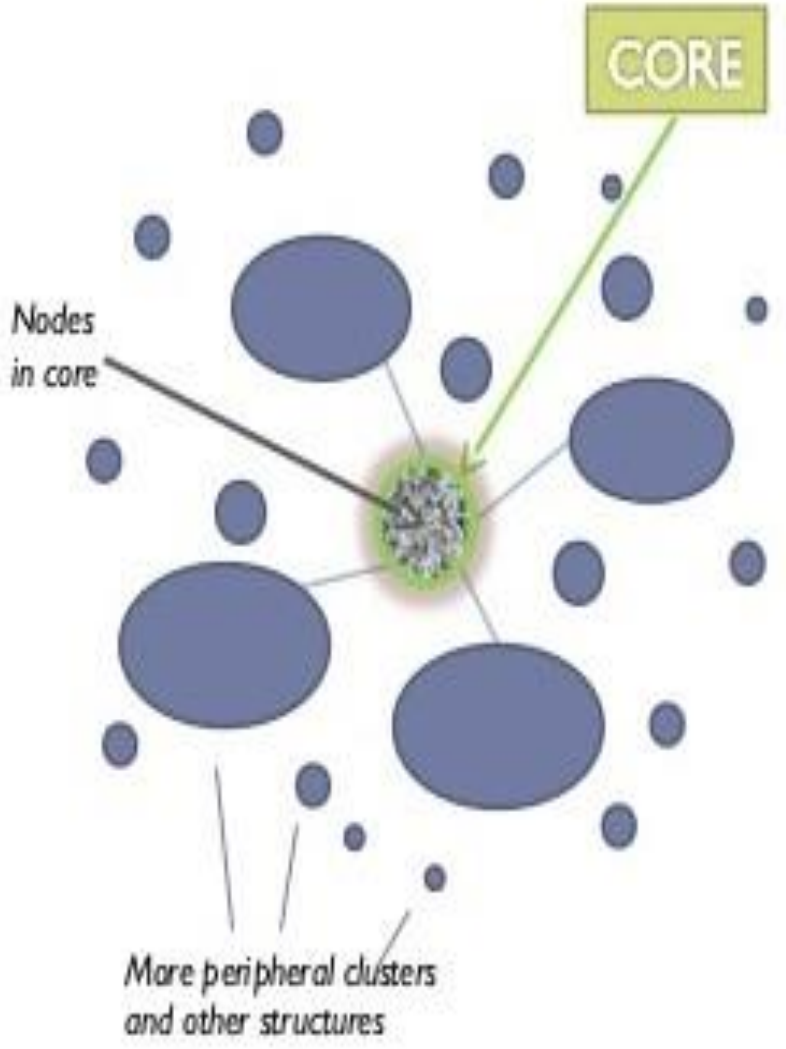
- based on the differences in degrees between nodes
  (and usually normalized to values between 0 and 1)
- a network that greatly depends on 1-2 highly connected node (e.g., as a result of preferential attachment) exhibits greater differences in degree centrality between nodes
- centralized structures can perform better at some tasks, like team-based problem-solving, that require coordination
  - × but are more prone to failure if key players disconnect



Core-Periphery Structures

Centralization measure *(continued)*

- many large groups/communities have a ***core*** of densely interconnected users
  - this is critical for connecting a much larger periphery
  - the cores can be identified visually, or by examining the location of high-degree nodes and their joint degree distributions
  - ***bow-tie analysis*** *(used to analyze the structure of the Web)* can also be used to distinguish between the core and other, more peripheral elements in a network



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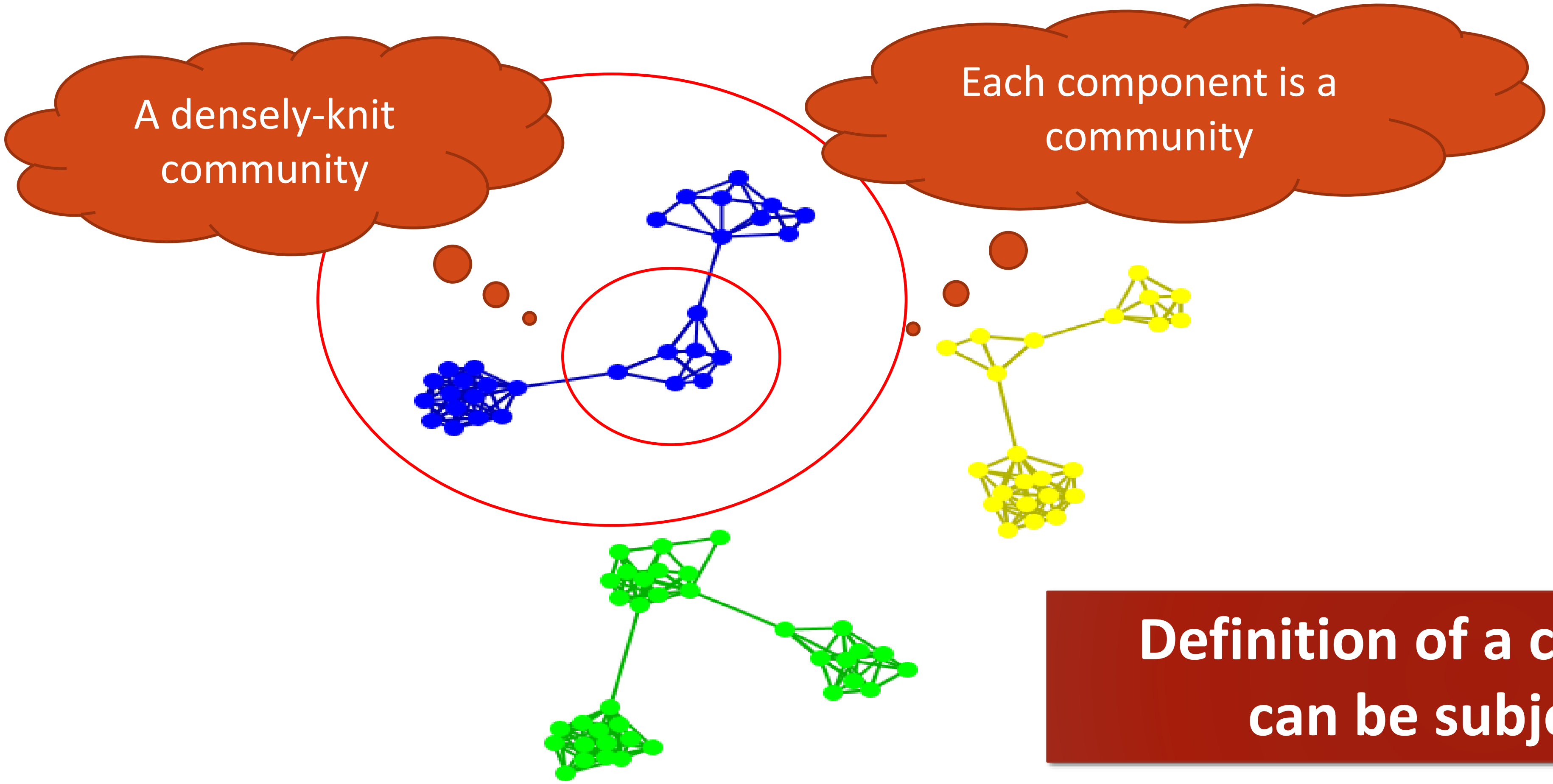
Communities in Social Networks

- *Communities* are formed by individuals such that those within a group interact with each other more frequently than with those outside the group
  - a.k.a. *group, cluster, cohesive subgroup, module* in different contexts
- *Community detection:* discovering groups in a network where individuals' group memberships are not explicitly given
- Why communities in social media?
  - Human beings are social
  - Easy-to-use social media allows people to extend their social life
  - Difficult to meet friends in the physical world, but much easier to find friends online
  - Interactions between nodes can help determine communities

Communities in Social Media

- Two types of groups in social media
  - *Explicit Groups:* formed by user subscriptions
  - *Implicit Groups:* implicitly formed by social interactions
- Some social media sites allow people to join groups. Is it necessary to extract groups based on network topology?
  - Not all sites provide a community platform
  - Not all people want to make an effort to join groups
  - Groups can change dynamically
- Network interaction provides information about the relationship between users
  - Can complement other kinds of information
  - Help network visualization and navigation
  - Provide basic information for other tasks

Subjectivity of Community Definition



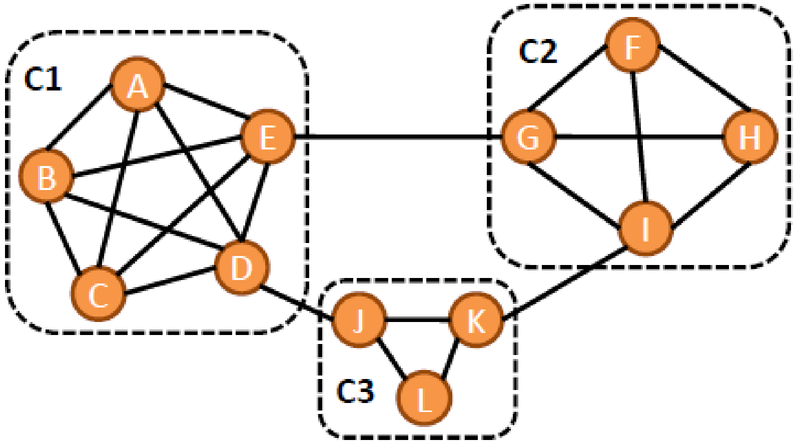
Definition of a community can be subjective.

Community Detection:
Communities, Modules, or Clusters

What are network communities?

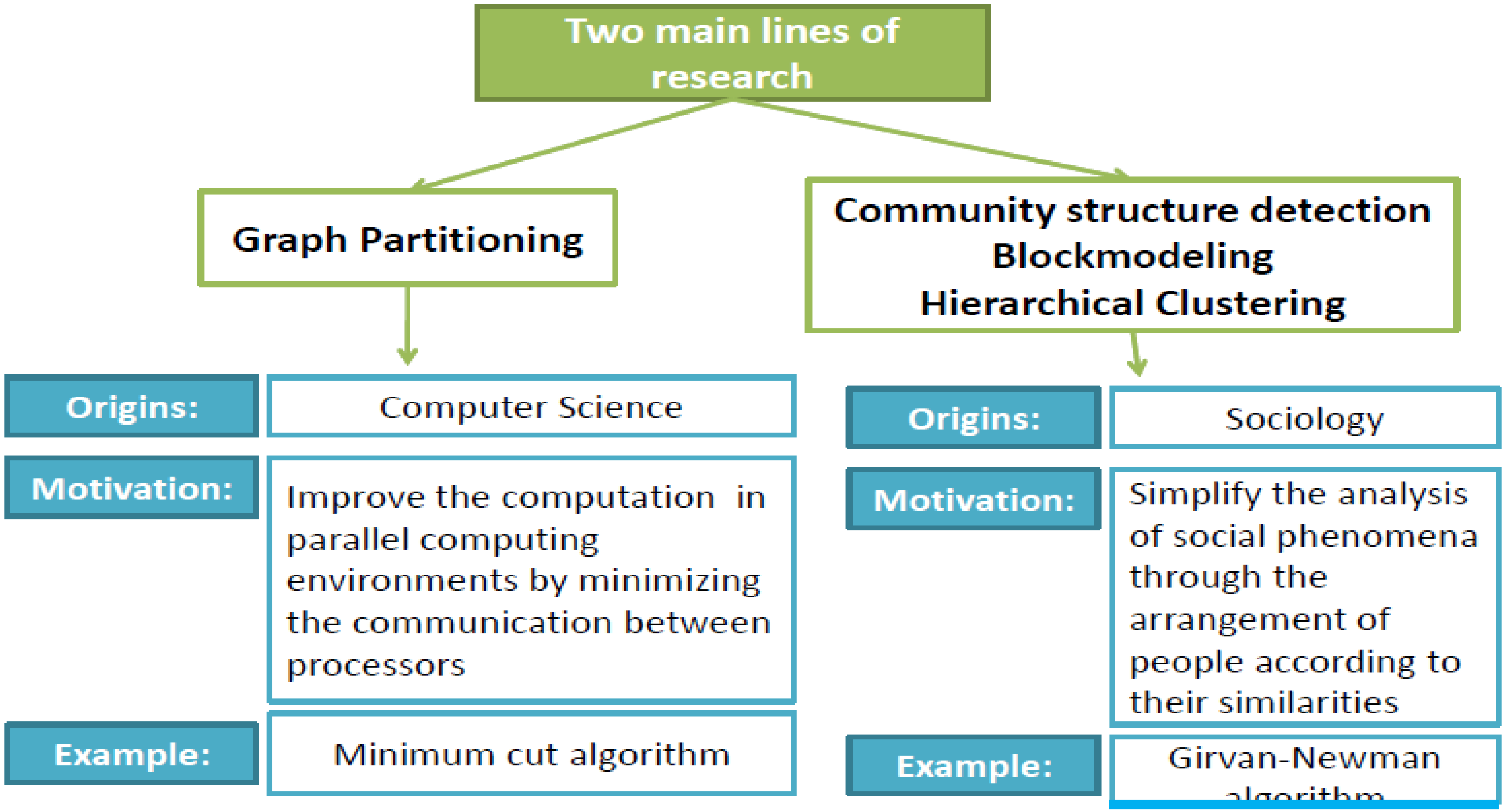
- Groups of similar nodes
- Groups of densely interconnected vertices, with sparser connections between these groups

Examples: families, work groups, circles of friends, topic-related web pages, customers of a given product, etc.



This graph has 3 communities:
C1={A,B,C,D,E}
C2={F,G,H,I}
C3={J,K,L}

Community Detection



Community Detection

- **Methods to address the problem of finding communities in graph data:**
  - **Minimum cut**
  - Hierarchical clustering using ***cosine similarity, Jaccard index, Euclidean distance, and Hamming distance*** as measures of similarity
  - **Girvan-Newman algorithm**
  - Walktrap – runs short random walks and uses their results to merge separate communities in a bottom-up manner
  - **Modularity Optimization** using, e.g., ***greedy techniques, simulated annealing***
  - Clique percolation method for finding overlapping communities
  - Block modeling and other methods

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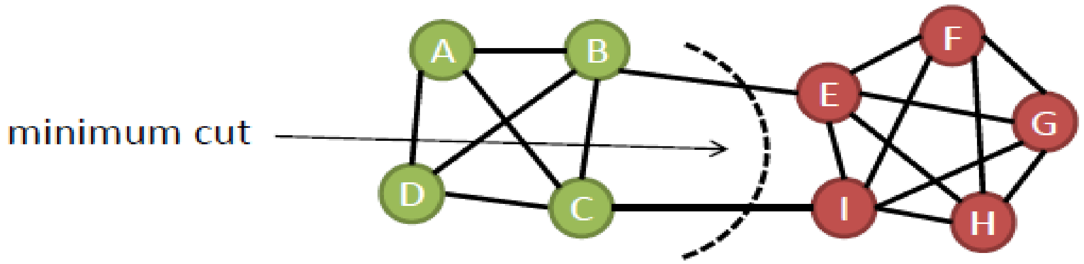
Community Detection: Minimum-Cut Algorithm

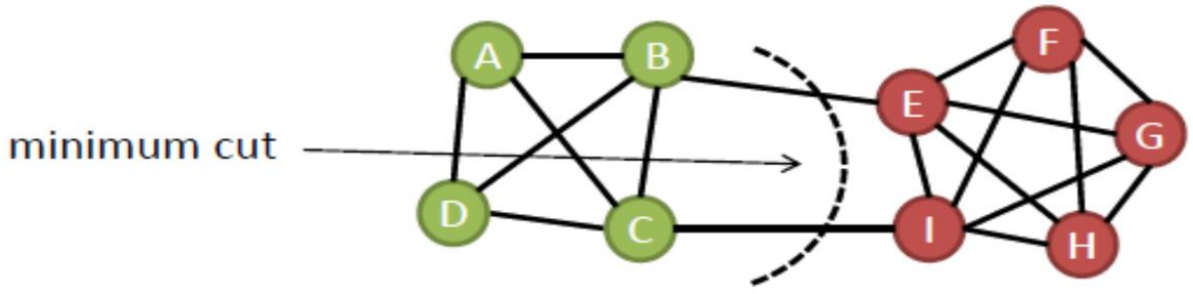


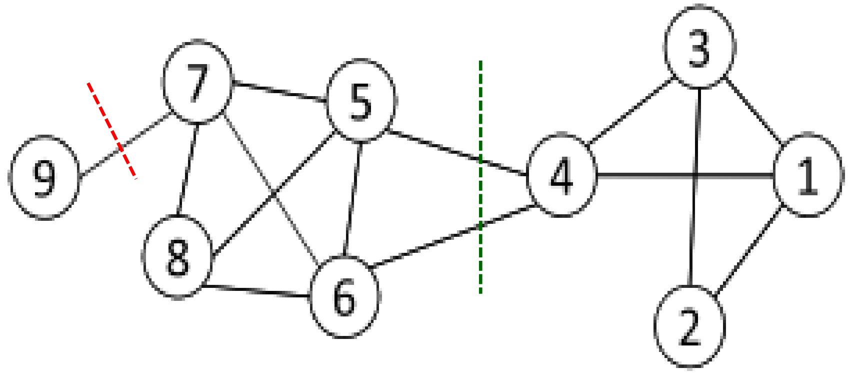
Minimum-Cut (Ahuja et al., 1993)

- **Idea:** divide the network into two communities by minimizing the number of edges (or the sum of the weights, in weighted networks) running through unlike groups, also known as the cut-size
- **For k>2:** implement the strategy of iterative bisecting (in the second and further iterations, groups are sequentially divided into two sub-groups)
- **Drawback:** produces groups of unbalanced sizes
- **Computational complexity:**
  - a simple solution uses the Max-Flow based s-t cut algorithm with the complexity $O(n^5)$
  - an implementation using Karger´s algorithm has the time complexity $O(n^4)$, but it is not guaranteed always to find the minimum cut
  - for $w_{ij}=1$ and no balancing constraints on the partitions, the 2-way cut problem is polynomially solvable (Karger´s randomized minimum cut) in $O(n^2 \log n)$
  - arbitrary edge weights and balancing constraints make the problem NP-hard

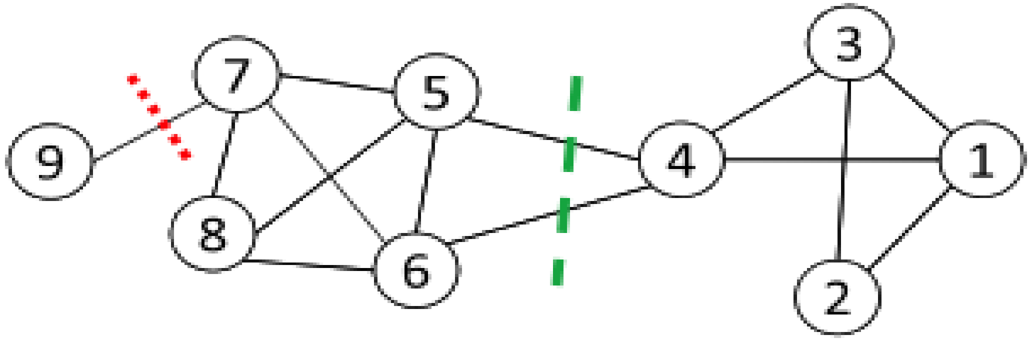
Minimum Cut

- **Most interactions occur within a group; interactions between groups are rare**
  - **community detection → minimum cut problem**
- **Cut:** A partition of the vertices of a graph into two disjoint sets
- **Minimum cut problem:** find a graph partition such that the number of edges between the two sets is minimized





Ratio Cut and Normalized Cut



Minimum cut often returns an imbalanced partition, with one set being a singleton, e.g., node 9

➔ **Change the objective function to consider community size**

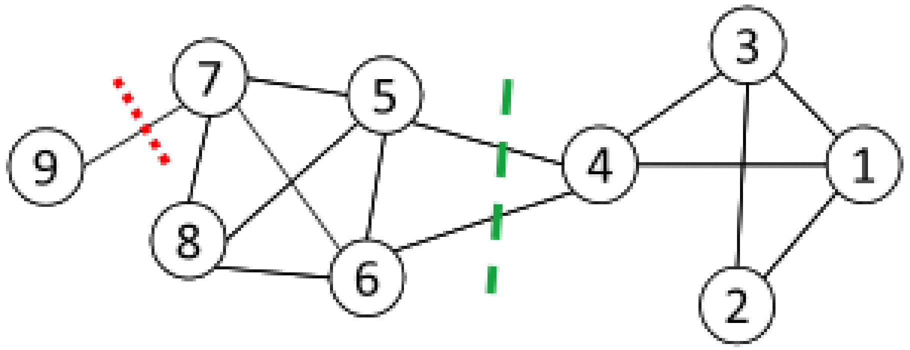
$$\text{Ratio Cut}(\pi) = \frac{1}{k}\sum_{i=1}^{k}\frac{cut(C_i, \bar{C}_i)}{|C_i|},$$

$$\text{Normalized Cut}(\pi) = \frac{1}{k}\sum_{i=1}^{k}\frac{cut(C_i, \bar{C}_i)}{vol(C_i)}$$

π: the partition; $C_{i,}$: a community
$|C_i|$: number of nodes in C_i
$vol(C_i)$: sum of degrees in C_i
$cut(A,B) = \sum_{i \in A, j \in B} e_{ij}$

Ratio Cut and Normalized Cut: an Example

For partition in red: π_1



$$\text{Ratio Cut}(\pi_1) = \frac{1}{2}\left(\frac{1}{1} + \frac{1}{8}\right) = 9/16 = 0.56$$

$$\text{Normalized Cut}(\pi_1) = \frac{1}{2}\left(\frac{1}{1} + \frac{1}{27}\right) = 14/27 = 0.52$$

For partition in green: π_2

$$\text{Ratio Cut}(\pi_2) = \frac{1}{2}\left(\frac{2}{4} + \frac{2}{5}\right) = 9/20 = 0.45 < \text{Ratio Cut}(\pi_1)$$

$$\text{Normalized Cut}(\pi_2) = \frac{1}{2}\left(\frac{2}{12} + \frac{2}{16}\right) = 7/48 = 0.15 < \text{Normalized Cut}(\pi_1)$$

Both ratio cut and normalized cut prefer a balanced partition

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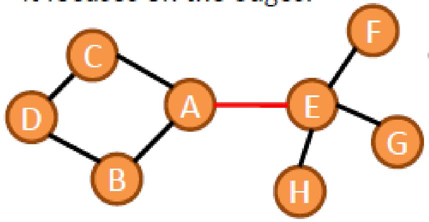
Community Detection: Girvan-Newman Algorithm

Edge betweenness:

The idea is the same but instead of focusing on nodes, it focuses on the edges.



$$b_e = \sum_{u,v \in V(G)} \frac{\sigma_{uv}(e)}{\sigma_{uv}}$$

the total number of shortest paths between the vertices u and v

Girvan-Newman algorithm (Girvan and Newman, 2002)

~ a divisive hierarchical algorithm, based on a top-down approach

- decomposes the initial full graph into progressively smaller connected pieces, until there are no edges to remove, and each node represents itself a community

- **Idea:** identify edges that connect vertices belonging to different communities (the so-called ***bridges***) and, iteratively, remove them from the graph
- **Criterion:** ***edge betweenness*** is adopted to identify and delete bridges (i.e., edges with a high betweenness), since these edges lie on a large number of shortest paths between vertices
- **Drawback:** suitable only for networks of a moderate size due to high computational costs - $O(m^2n)$ for graphs having m edges and n vertices, or $O(n^3)$ for sparse networks

Community Detection: Girvan-Newman Algorithm

Girvan-Newman algorithm (Girvan and Newman, 2002):

Input: the entire graph

Output: hierarchical structure to be represented, e.g., as a *dendrogram*

1. Compute the betweenness of all edges in the network
2. Remove the edge with the highest betweenness
3. Repeat the previous steps until there are no edges to remove in the graph

- **Selecting the number of communities**
  - The algorithm returns a set of possible solutions; however, it does not indicate which solution is the best.
  - To select the best partition, compute the modularity of each network's division. Then choose the partition with the highest modularity.

Community Detection:
Evaluating Community Quality

Modularity Q ~ quantifies the quality of a given graph division into communities

- **Basic Idea:** a network has a meaningful community structure if the number of edges between communities is smaller than expected based on a random choice
- **Modularity values:** can be either positive or negative; Q ϵ [-1,1].
  - If positive, then there is a possibility of finding community structure on the network
  - If the values are large and positive, the corresponding partition may reflect the true community structure. For meaningful communities, modularity Q should be >=0.3 (Clauset et al.,2004).

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j),$$

$\frac{k_i k_j}{2m}$ - expected number of edges falling between vertices ***i*** and ***j***

$\delta(c_i,c_j)$ - Kronecker delta

A_{ij} - entry of the adjacency matrix that gives the number of edges between vertices ***I*** and ***j***

m – number of edges k_i – degree of vertex ***i***

c_i – group to which vertex ***i*** belongs

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Community Detection: Louvain Algorithm

Louvain method (Blondel et al., 2008):

~ a greedy optimization method for agglomerative hierarchical modularity optimization

- **Process:** the algorithm comprises two phases
  1. look for local optima by maximizing modularity in a local way
  2. then, aggregate nodes belonging to the same community and create a new network where each node represents one of the previously found communities
- **Advantages:** achieves *good performance* also in large network graphs at *low computational costs* – $O(n \log n)$, n is the number of vertices
- **Drawback:** the results are order-sensitive

Community Detection: Louvain Algorithm

Phase 1: consider each node as a single community

1.1. Compute modularity Q

1.2. Move the isolated node from its community to a neighboring community

1.3. Compute the gain/loss in modularity yielded by the changed assignment

1.4. If the modularity increases (~ there is a gain), keep the node in the "new" community

1.5. Repeat the process until no further improvements are possible

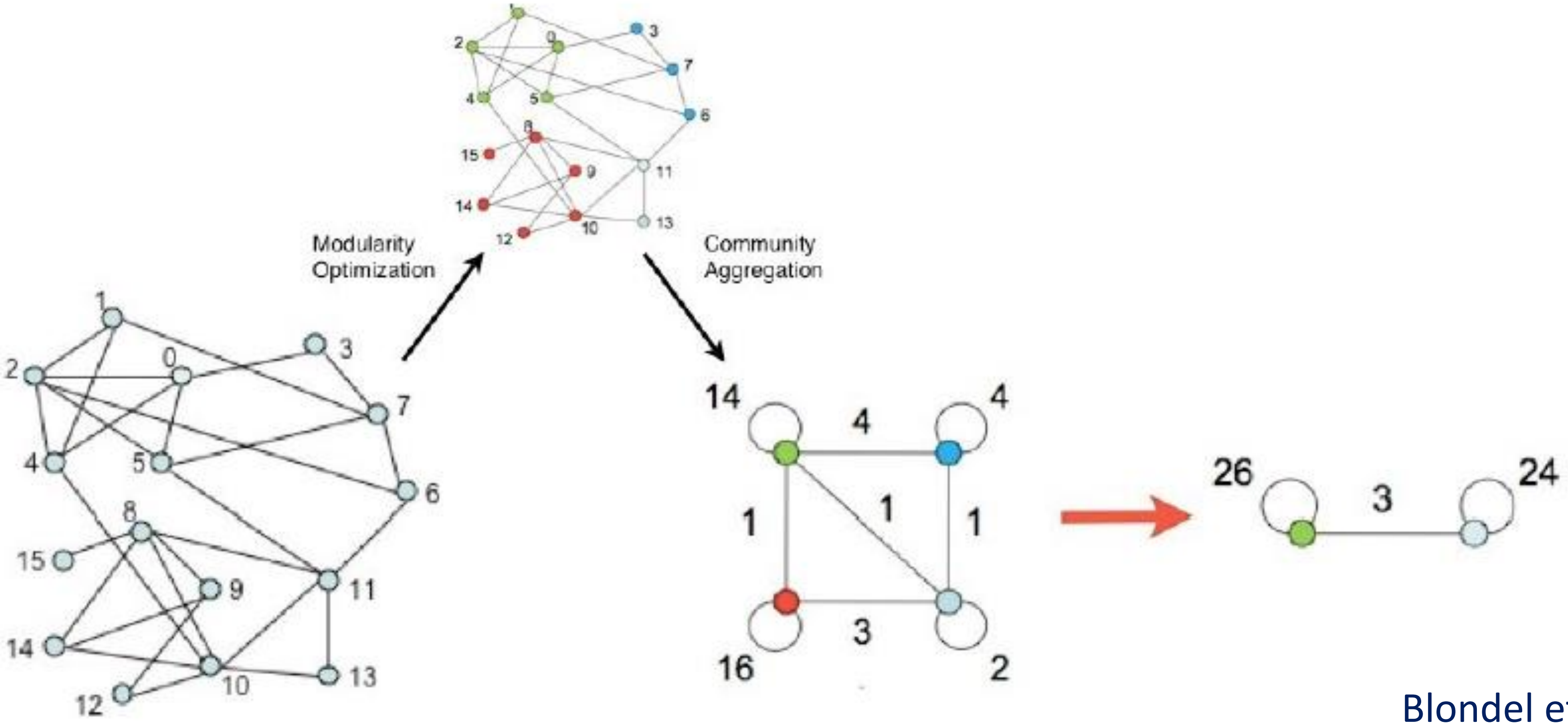
Output: the k-th level partition (k is the number of the iteration)

Phase 2: create a new network (~ supergraph), derived from the original one, where each new node (~ supervertex) is the aggregation of the nodes assigned to a given community (found in Phase 1); two supervertices are connected by an edge if there is at least one edge linking two vertices inside the corresponding community.

Iteration: Repeat Phases 1 and 2 until the maximum modularity is attained

Community Detection: Louvain Method

Illustration of the process behind the algorithm:

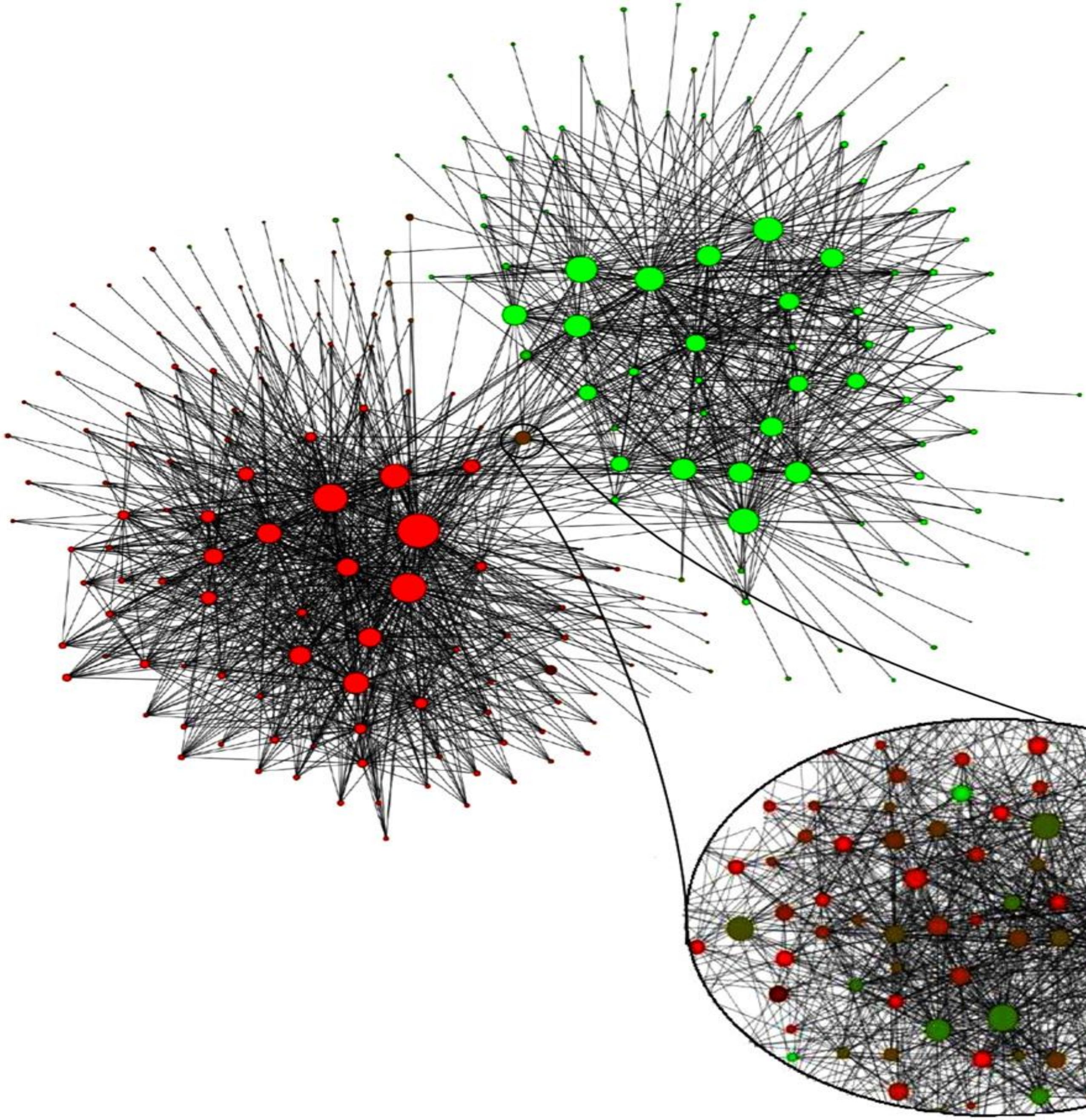


Blondel et al., 2008

An Application of the Louvain Method:

Network Analysis of a Belgian Mobile Phone Operator

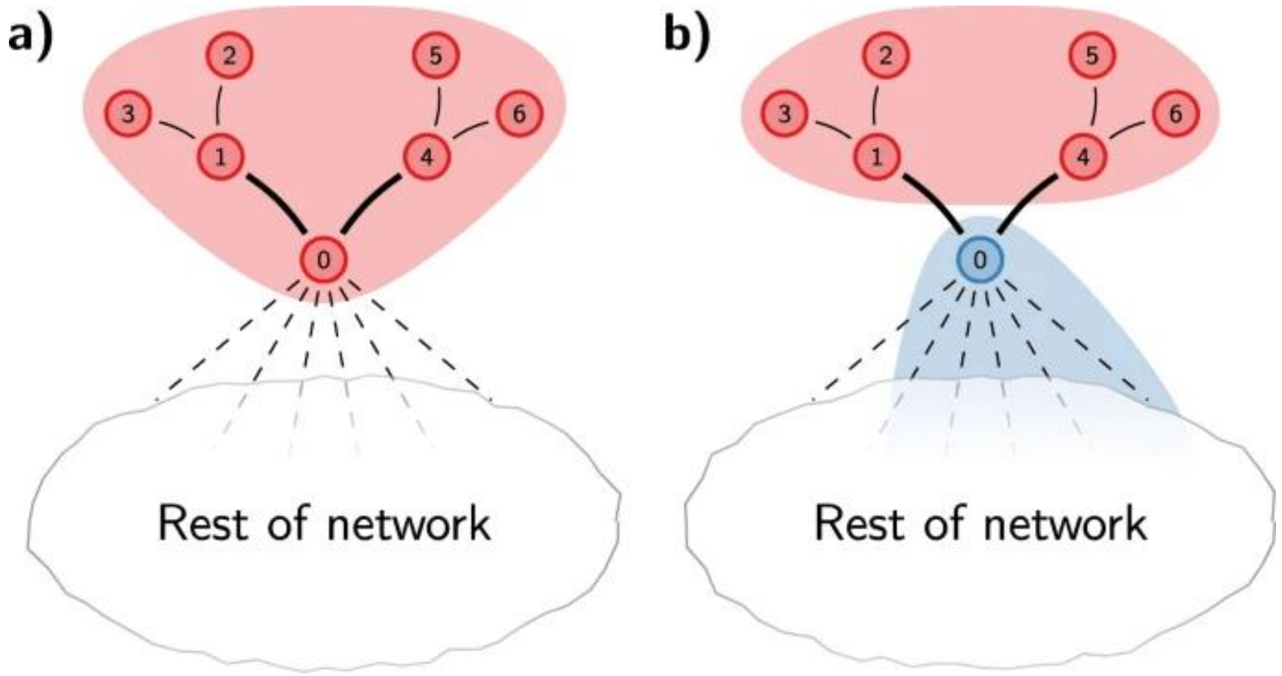
- cca 2 million customers
- size of a node is proportional to the number of individuals in the community
- color represents the main language spoken in the community (red for French and green for Dutch)
- communities of more than 100 customers plotted only
- A zoom at higher resolution reveals the intermediate community made of sub-communities with less apparent language separation



Community Detection: Louvain Method

Disadvantages:

- Disconnected communities
- Getting stuck in local optima
- Failing to refine communities after initial merging



Right: A graph illustrating how communities can become disconnected when using the Louvain algorithm.

Consider the partition shown in (a). When node 0 is moved to a different community, the red community becomes internally disconnected, as shown in (b). However, nodes 1–6 are still locally optimally assigned, and will therefore remain in the red community.

Community Detection: Leiden Algorithm

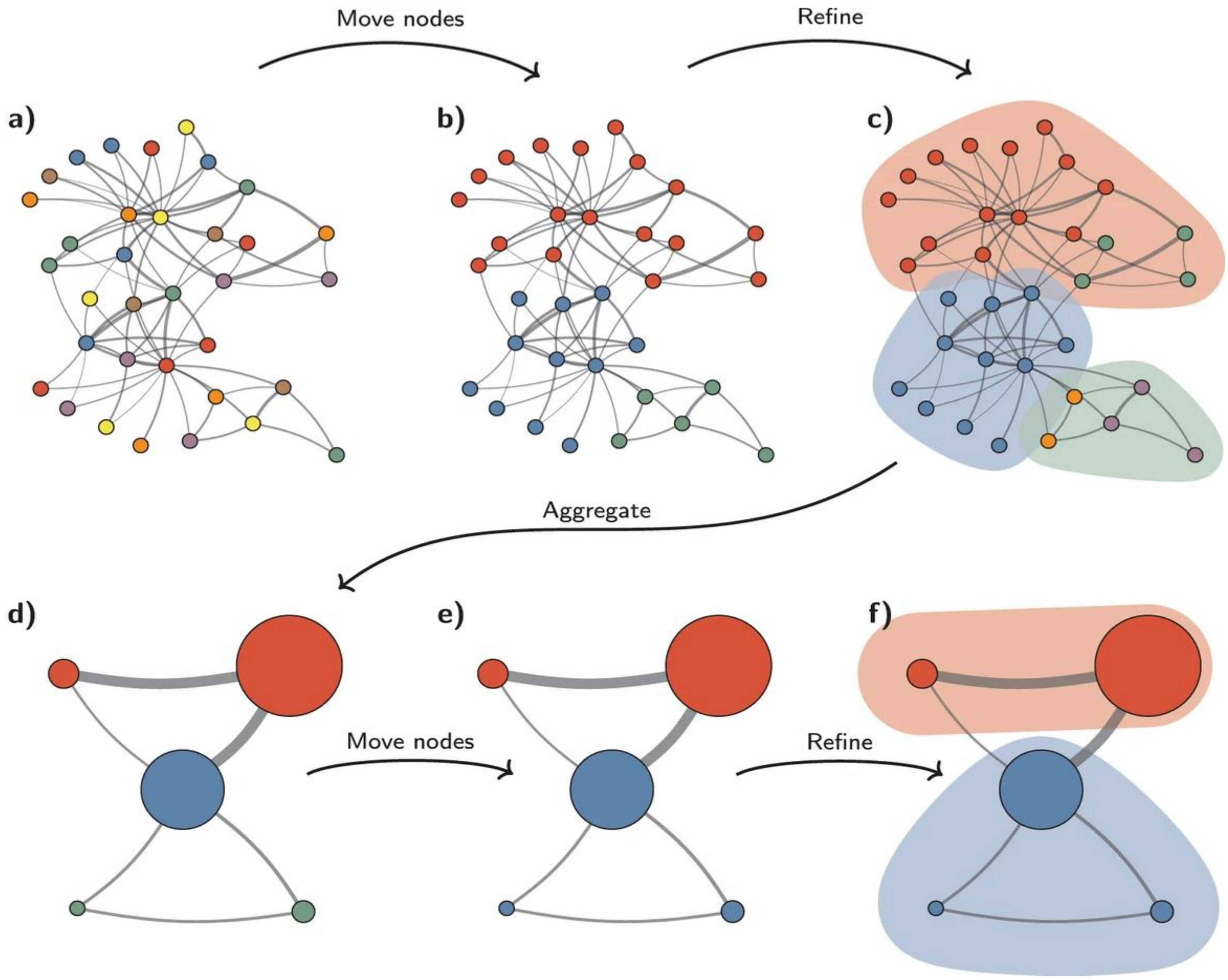
Leiden algorithm (Traag, Waltman, and van Eck, 2019):

~ a refinement of the Louvain method for agglomerative hierarchical modularity optimization

- **Process:** the algorithm comprises three phases
  1. local moving of nodes (similar to the Louvain method),
  2. refinement of the partition,
  3. and network aggregation (similar to the Louvain method).
- **Advantages:** the Leiden algorithm is more efficient than the Louvain method, but may lead to extended processing times for massive graphs; parallel multicore implementations may boost the speed
- **Drawback:** hard partition (the nodes can belong to only one community)

Community Detection:
the Leiden Algorithm

A graph depicting the steps of the Leiden algorithm:



Community Detection: Challenges

Despite many algorithms, community detection is still considered a challenging problem, due to 2 main reasons:

1. Typically, the ***number of communities*** is unknown and has to be determined
2. Communities are usually ***heterogeneous*** in terms of size and density.

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Other Community Detection Techniques

Roughly, community detection methods can be divided into 4 categories, although their criteria may vary depending on the task:

- Node-Centric Community
  - Each node in a group satisfies certain properties
- Group-Centric Community
  - Consider the connections within a group as a whole. The group has to satisfy certain properties without zooming into the node-level
- Network-Centric Community
  - Partition the whole network into several disjoint sets
- Hierarchy-Centric Community
  - Construct a hierarchical structure of communities

Node-Centric Community Detection

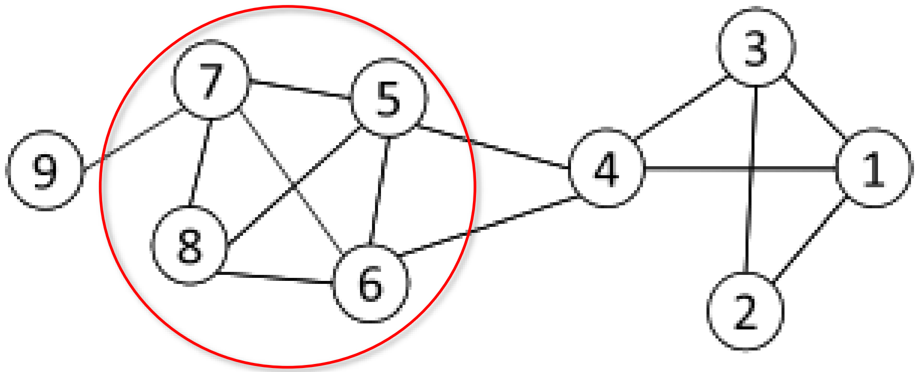
Nodes may be required to satisfy different properties commonly used in traditional social network analysis:

- Complete Mutuality - cliques
- Reachability of members - k-clique, k-clan, k-club
- Nodal degrees - k-plex, k-core
- Relative frequency of Within-Outside Ties - LS sets, Lambda sets

Here, we will discuss some of the representative properties only

Complete Mutuality: Cliques

Clique: a maximum complete subgraph in which all nodes are adjacent to each other



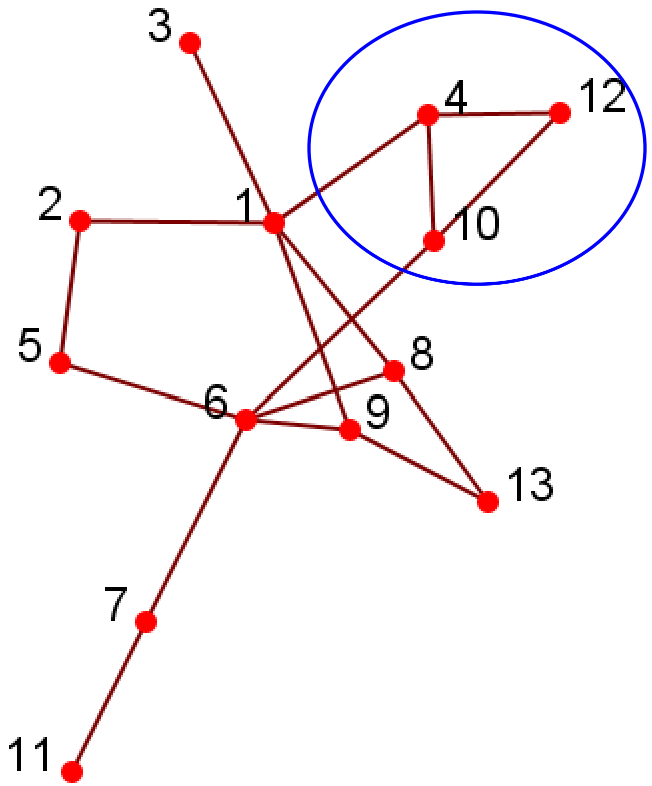
Nodes 5, 6, 7, and 8 form a clique

NP-hard to find the maximum clique in a network!

Straightforward implementation to find cliques is very expensive in terms of time complexity

Complete Mutuality: Clique

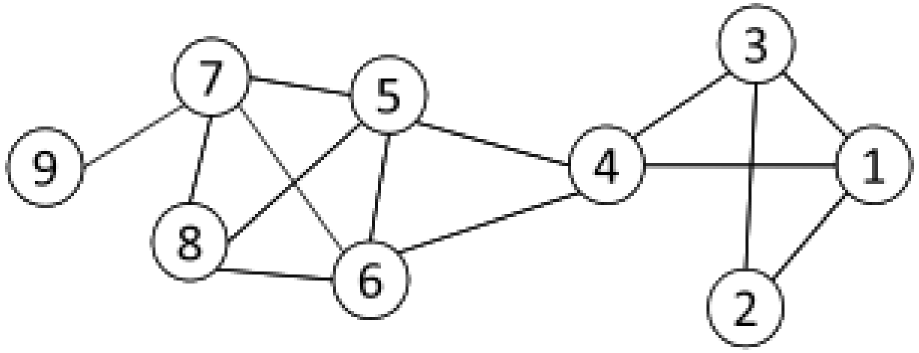
- Clique: a maximal complete subgraph of three or more nodes, all of which are adjacent to each other
- NP-hard to find the maximal clique
- The idea of **recursive pruning**: To find a clique of size k, remove those nodes with less than k-1 degrees
- Normally, use cliques as a core or seed to explore larger communities

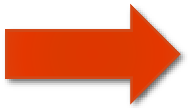


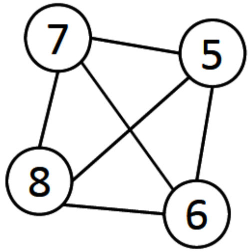
Finding the Maximum Clique

- In a clique of size k, each node maintains a degree $\geq k-1$
- Nodes with degree $< k-1$ will not be included in the maximum clique
- **Recursively apply the following pruning procedure**
  - Sample a sub-network from the given network, and find a clique in the sub-network, say, by a greedy approach
  - Suppose the clique above is size k, to find a *larger* clique, all nodes with degree $\leq k-1$ should be removed.
- Repeat until the network is small enough
- Many nodes will be pruned as social media networks follow a power law distribution for node degrees

Maximum Clique Example







Suppose we sample a sub-network with nodes {1-5} and find a clique {1, 2, 3} of size 3

In order to find a clique >3, remove all nodes with degree <=3-1=2

- Remove nodes 2 and 9
- Remove nodes 1 and 3
- Remove node 4

Clique Percolation Method (CPM)

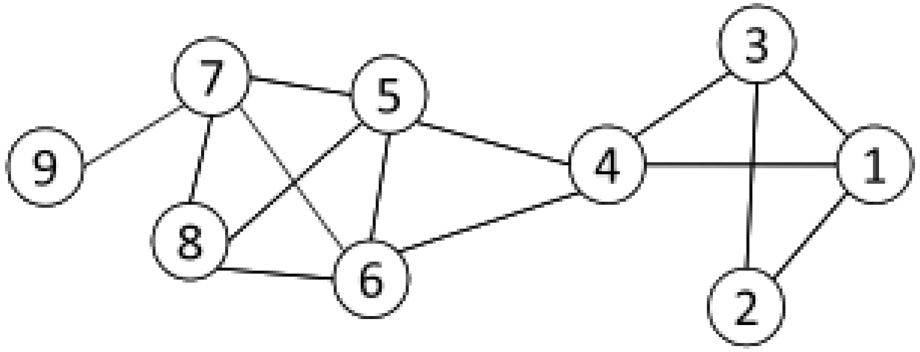
Clique is a stringent definition, unstable

➔ in practice, use cliques as **a core or a seed** to find larger communities

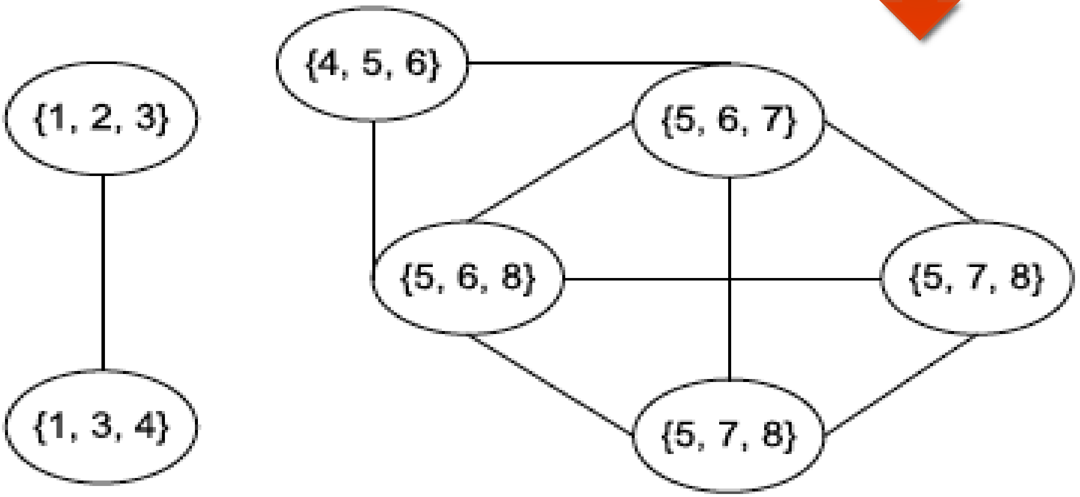
CPM is an example of a method to find **overlapping communities**

- **Input**
  - A parameter k, and a network
- **Procedure**
  - Find out all cliques of size k in a given network
  - Construct a clique graph. Two cliques are adjacent if they share $k-1$ nodes
  - Each connected components in the clique graph form a community

Clique Percolation Method (CMP) – an example



Cliques of size 3:
{1, 2, 3}, {1, 3, 4}, {4, 5, 6}, {5, 6, 7}, {5, 6, 8}, {5, 7, 8}, {6, 7, 8}



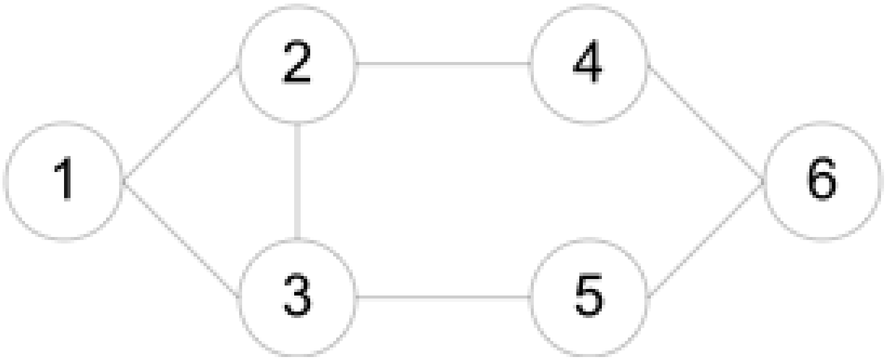
Communities:
{1, 2, 3, 4}
{4, 5, 6, 7, 8}

Reachability: k-clique, k-club

The main idea: any node in a group should be reachable in k hops

k-clique: a maximal subgraph with the largest geodesic distance between any nodes $\leq k$

k-club: a substructure of diameter $\leq k$

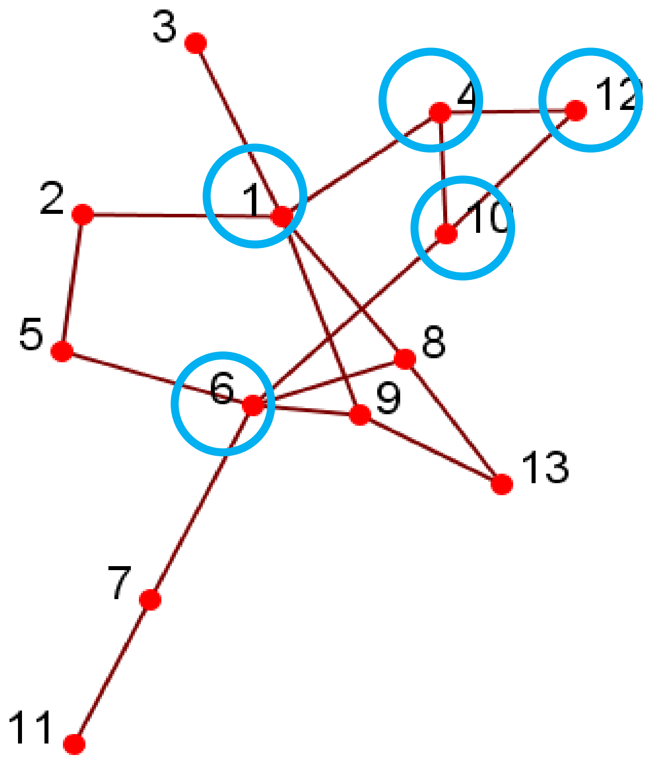


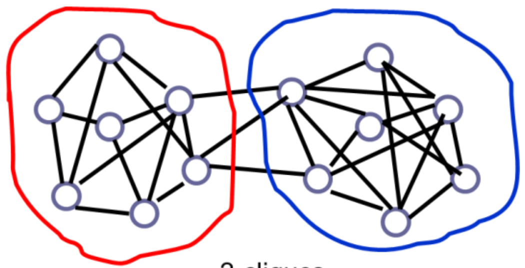
Cliques: {1, 2, 3}
2-cliques: {1, 2, 3, 4, 5}, {2, 3, 4, 5, 6}
2-clubs: {1,2,3,4}, {1, 2, 3, 5}, {2, 3, 4, 5, 6}

Commonly used in traditional SNA and often involves combinatorial optimization

Reachability: k-clique, k-club: Example 2

- Any node in a group should be reachable in k hops
- ***k*-clique:** a maximum subgraph in which the largest geodesic distance between any nodes $\leq k$
  - **a k-clique can have diameter >= *k* within the subgraph** (e.g., the 2-clique {12, 4, 10, 1, 6}; the diameter within the subgraph, d(1, 6) = 3)
- ***k*-club:** a substructure of diameter <=*k*
  - e.g., {1,2,5,6,8,9}, {12, 4, 10, 1} are 2-clubs
- ***k*-clan:** a *k*-clique with the diameter <=*k*,
  - e.g., {1,2,3,4,5} is a 3-clan
- node-centric community definitions are either too strict, or it is too time-consuming to find the groups





2-cliques

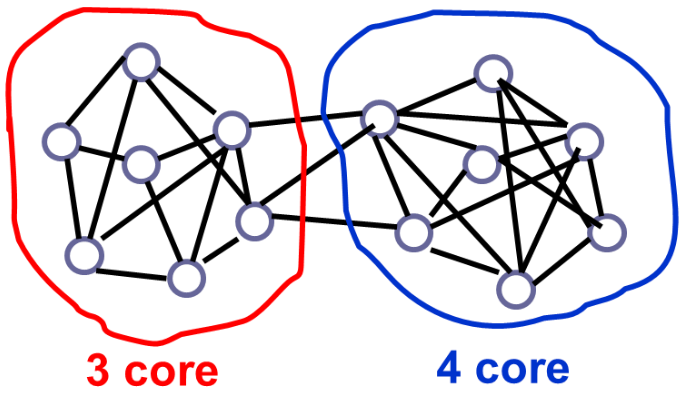
Nodal Degrees: k-core, k-plex

Each node should have a certain number of connections to nodes within the group

- **k-core:** a substructure where each node is connected to at least k other members within the group
- **k-plex:** for a group with n_s nodes, each node should be adjacent to no fewer than n_s-k in the group

The definitions are complementary:

- a k-core is a $(n_s\text{-}k)$-plex
- however, the set of all k-cores for a given k is not identical to the set of all $(n_s\text{-}k)$-plexes, as the group size n_s can vary across k-cores.



Within-Outside Ties: Luccio-Sami sets (LS sets)

LS sets: any of the proper subsets of an LS set has more ties to other nodes in the group than outside of the group

- **Objective:** find cohesive groups of nodes for circuit design on chips (Luccio and Sami)
- Too strict, not reasonable for network analysis

A relaxed definition is a ***k*-component**

- A maximum subgraph that remains connected even after removing fewer than *k* vertices or edges
- Requires the computation of edge-connectivity between any pair of nodes via the minimum-cut, maximum-flow algorithm

Recap of Node-Centric Communities

Each node has to satisfy specific properties

- Complete mutuality
- Reachability
- Nodal degrees
- Within-Outside ties

Limitations:

- Too strict, but can be used as the core of a community
- Not scalable, commonly used in network analysis with small-sized networks
- Sometimes, it is not consistent with the properties of large-scale networks
  - e.g., node degrees in scale-free networks

Group-Centric Community Detection: Density-Based Groups

Group-centric criteria require the whole group to satisfy a particular condition

- e.g., the group density is to be big enough (>= a given threshold)

A subgraph $G_S (V_S , E_S)$ is a γ-dense quasi-clique if

$$\frac{|E_s|}{|V_s|(|V_s|-1)/2} \geq \gamma$$

<u>A similar strategy to that of cliques can be used:</u>

- Local search: Sample a subgraph, and find a maximum γ-dense quasi-clique (say, of size k)
- Heuristic pruning: Remove nodes with degree less than the average degree, ($< (k - 1) \gamma$)

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Network-Centric Community Detection

Network-centric criteria consider connections within a network globally

The goal: partition the nodes of a network into disjoint sets

Approaches:

- Clustering based on vertex similarity
- Latent space models
- Block model approximation
- Spectral clustering
- Modularity maximization

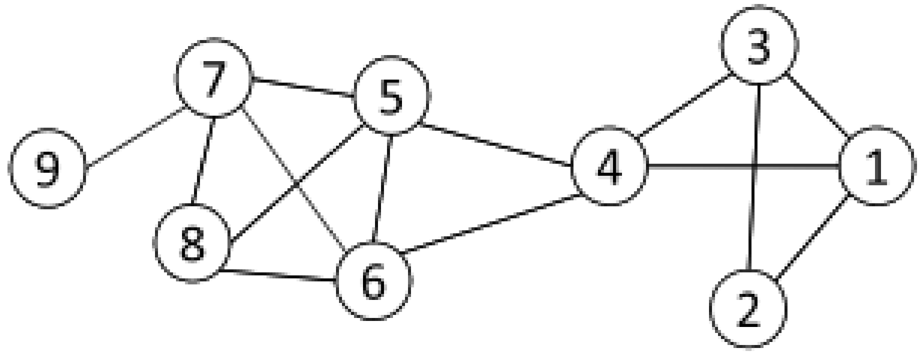
Clustering Based on Vertex Similarity

Apply k-means or similarity-based clustering to nodes

Vertex similarity is defined in terms of the similarity of their neighborhood

Structural equivalence: two nodes are structurally equivalent iff they are connected to the same set of actors

Nodes 1 and 3 are structurally equivalent; So are nodes 5 and 6.

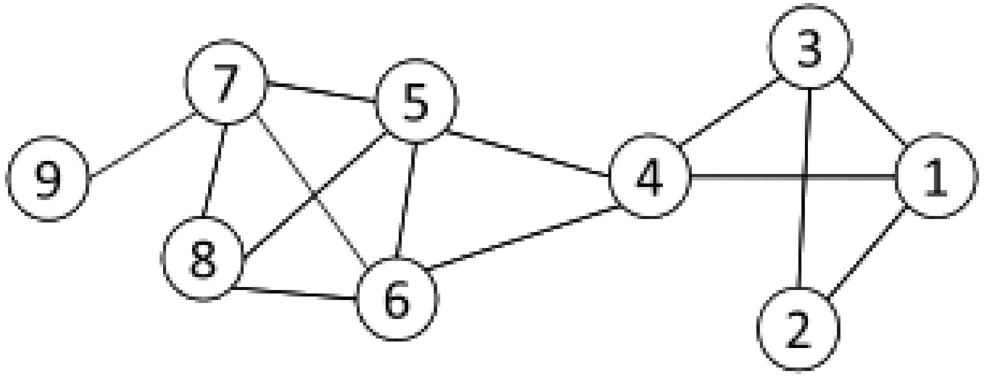


Structural equivalence is too restrictive for practical use.

Vertex Similarity

Jaccard Similarity:

$$Jaccard(v_i, v_j) = \frac{|N_i \cap Nj|}{|N_i \cup Nj|}$$



$$Jaccard(4,6) = \frac{|\{5\}|}{|\{1,3,4,5,6,7,8\}|} = \frac{1}{7}$$

Cosine similarity:

$$cosine\,(v_i, v_j) = \frac{\sum_k A_{ik}A_{jk}}{\sqrt{\sum_s A_{is}^2 \cdot \sum_t A_{jt}^2}}$$

$$cosine(4,6) = \frac{1}{\sqrt{4 \cdot 4}} = \frac{1}{4}$$

Clustering Based on Vertex Similarity

For practical use with huge networks (consider the connections as features):

- use *Cosine* or *Jaccard similarity* to compute vertex similarity
- apply the classical k-means clustering Algorithm

k-means Clustering Algorithm

- each cluster is associated with a centroid (center point)
- each node is assigned to the cluster with the closest centroid

Algorithm 1 Basic K-means Algorithm.

1: Select K points as the initial centroids.
2: **repeat**
3: Form K clusters by assigning all points to the closest centroid.
4: Recompute the centroid of each cluster.
5: **until** The centroids don't change

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Hadamard product (~ element-wise product): $(A \circ B)_{ij} = (A)_{ij}(B)_{ij}$ for the matrices A and B of the same dimension.

Latent Space Models

Map the nodes into a low-dimensional space such that the proximity between nodes based on network connectivity is preserved, then apply k-means clustering

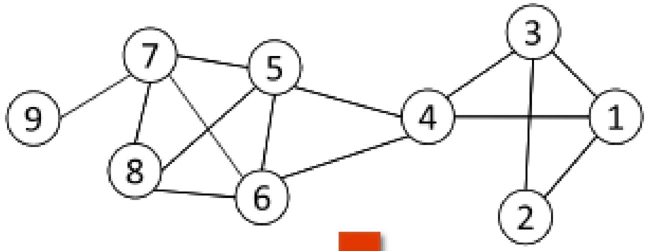
Multi-dimensional scaling (MDS)

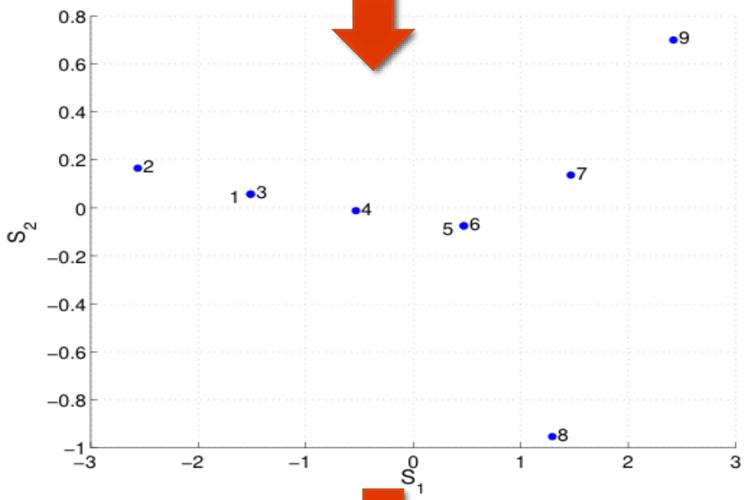
- Given a network, construct a proximity matrix *P* representing the pairwise distance between nodes (e.g., geodesic distance)
- Let $S \in R^{nxl}$ denote the coordinates of nodes in the low-dimensional space

$$SS^T \approx -\frac{1}{2}(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T)(P \circ P)(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T) = \widetilde{P}$$

- **Objective function:** $\min \|SS^T - \widetilde{P}\|_F^2$
- **Solution:** $S = V\Lambda^{\frac{1}{2}}$
- *V* is the top *l* eigenvectors of $\widetilde{P}$, and Λ is a diagonal matrix of top eigenvalues $\Lambda = diag(\lambda_1, \lambda_2, \cdots, \lambda_\ell)$

Multi-Dimensional Scaling (MDS): an Example





Two communities: {1, 2, 3, 4} and {5, 6, 7, 8, 9}

geodesic distance

$$P = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 3 & 3 & 4 \\ 1 & 0 & 1 & 2 & 3 & 3 & 4 & 4 & 5 \\ 1 & 1 & 0 & 1 & 2 & 2 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 1 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 1 & 0 & 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 1 & 1 & 0 & 1 & 1 & 2 \\ 3 & 4 & 3 & 2 & 1 & 1 & 0 & 1 & 1 \\ 3 & 4 & 3 & 2 & 1 & 1 & 1 & 0 & 2 \\ 4 & 5 & 4 & 3 & 2 & 2 & 1 & 2 & 0 \end{bmatrix}$$

$$\widetilde{P} = \begin{bmatrix} 2.46 & 3.96 & 1.96 & 0.85 & -0.65 & -0.65 & -2.21 & -2.04 & -3.65 \\ 3.96 & 6.46 & 3.96 & 1.35 & -1.15 & -1.15 & -3.71 & -3.54 & -6.15 \\ 1.96 & 3.96 & 2.46 & 0.85 & -0.65 & -0.65 & -2.21 & -2.04 & -3.65 \\ 0.85 & 1.35 & 0.85 & 0.23 & -0.27 & -0.27 & -0.82 & -0.65 & -1.27 \\ -0.65 & -1.15 & -0.65 & -0.27 & 0.23 & -0.27 & 0.68 & 0.85 & 1.23 \\ -0.65 & -1.15 & -0.65 & -0.27 & -0.27 & 0.23 & 0.68 & 0.85 & 1.23 \\ -2.21 & -3.71 & -2.21 & -0.82 & 0.68 & 0.68 & 2.12 & 1.79 & 3.68 \\ -2.04 & -3.54 & -2.04 & -0.65 & 0.85 & 0.85 & 1.79 & 2.46 & 2.35 \\ -3.65 & -6.15 & -3.65 & -1.27 & 1.23 & 1.23 & 3.68 & 2.35 & 6.23 \end{bmatrix}$$

$$V = \begin{bmatrix} -0.33 & 0.05 \\ -0.55 & 0.14 \\ -0.33 & 0.05 \\ -0.11 & -0.01 \\ 0.10 & -0.06 \\ 0.10 & -0.06 \\ 0.32 & 0.11 \\ 0.28 & -0.79 \\ 0.52 & 0.58 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 21.56 & 0 \\ 0 & 1.46 \end{bmatrix}, \quad S = V\Lambda^{1/2} = \begin{bmatrix} -1.51 & 0.06 \\ -2.56 & 0.17 \\ -1.51 & 0.06 \\ -0.53 & -0.01 \\ 0.47 & -0.08 \\ 0.47 & -0.08 \\ 1.47 & 0.14 \\ 1.29 & -0.95 \\ 2.42 & 0.70 \end{bmatrix}$$

Block Models

Table 3.1: Adjacency Matrix

-	1	1	1	0	0	0	0	0
1	-	1	0	0	0	0	0	0
1	1	-	1	0	0	0	0	0
1	0	1	-	1	1	0	0	0
0	0	0	1	-	1	1	1	0
0	0	0	1	1	-	1	1	0
0	0	0	0	1	1	-	1	1
0	0	0	0	1	1	1	-	0
0	0	0	0	0	0	1	0	-

$$\min ||A - S\Sigma S^T||_F^2$$

Table 3.2: Ideal Block Structure

1	1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0	0
0	0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1	1

Σ is the so-called mixing matrix

S is the community indicator matrix

$$S = \begin{bmatrix} 0.20 & -0.52 \\ 0.11 & -0.43 \\ 0.20 & -0.52 \\ 0.38 & -0.30 \\ 0.47 & 0.15 \\ 0.47 & 0.15 \\ 0.41 & 0.28 \\ 0.38 & 0.24 \\ 0.12 & 0.11 \end{bmatrix}, \Sigma = \begin{bmatrix} 3.5 & 0 \\ 0 & 2.4 \end{bmatrix}.$$

Relax S to be numerical values, then the optimal solution corresponds to the top eigenvectors of A

Two communities: {1, 2, 3, 4} and {5, 6, 7, 8, 9}

Spectral Clustering

Both ratio cut and normalized cut can be reformulated as:

$$\min_{S \in \{0,1\}^{n \times k}} Tr(S^T \widetilde{L} S)$$

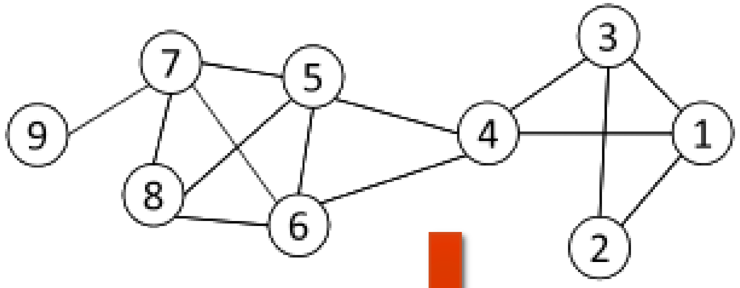
where $\widetilde{L} = \begin{cases} D - A & \text{graph Laplacian for ratio cut} \\ I - D^{-1/2} A D^{-1/2} & \text{normalized graph Laplacian} \end{cases}$

$D = diag(d_1, d_2, \cdots, d_n)$ a diagonal matrix of degrees

Spectral relaxation: $\min_S Tr(S^T \widetilde{L} S) \quad s.t. \; S^T S = I_k$

Optimal solution: top eigenvectors with the smallest eigenvalues

Spectral Clustering: an Example



$$D = diag(3,2,3,4,4,4,4,3,1)$$

$$\widetilde{L} = D - A = \begin{bmatrix} 3 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & 4 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 4 & -1 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 4 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 4 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 0.33 & -0.38 \\ 0.33 & -0.48 \\ 0.33 & -0.38 \\ 0.33 & -0.12 \\ 0.33 & 0.16 \\ 0.33 & 0.16 \\ 0.33 & 0.30 \\ 0.33 & 0.24 \\ 0.33 & 0.51 \end{bmatrix}$$

The 1st eigenvector means all nodes belong to the same cluster, no use

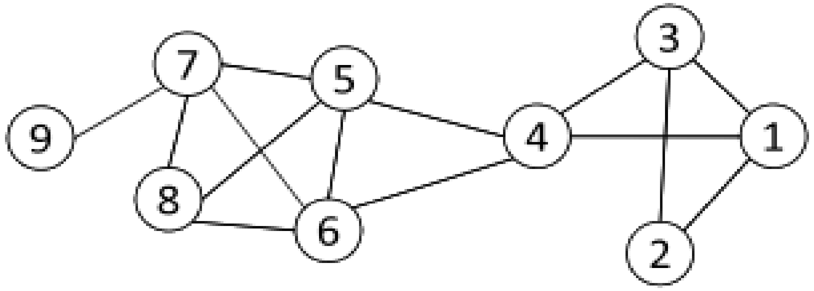
k-means

Two communities:
{1, 2, 3, 4} and {5, 6, 7, 8, 9}

Modularity Maximization

Modularity measures the strength of a community partition by taking into account the degree distribution

Given a network with m edges, the **expected number of edges** between two nodes with the degrees d_i and d_j is $\boldsymbol{d_i\, d_j / 2\, m}$



The expected number of edges between nodes 1 and 2 is: 3*2/ (2*14) = 3/14

Strength of a community: $\sum_{i\in C, j\in C}\left(A_{ij} - {d_i d_j}/{2m}\right)$

Modularity : $Q = \frac{1}{2m}\sum_{l=1}^{k}\sum_{i\in C_l, j\in C_l}\left(A_{ij} - {d_i d_j}/{2m}\right)$

A larger value indicates a good community structure.

Modularity Matrix

Modularity matrix: $B = A - \mathbf{d}\mathbf{d}^T/2m$ $(B_{ij} = A_{ij} - d_i d_j/2m)$

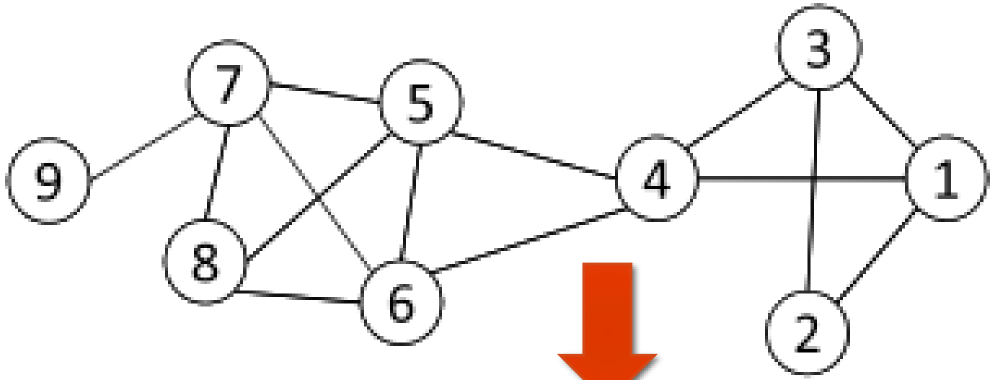
Similar to spectral clustering, Modularity maximization can be reformulated as

$$\max Q = \frac{1}{2m} Tr(S^T B S) \quad s.t.\ S^T S = I_k$$

Optimal solution: top eigenvectors of the modularity matrix

Apply *k*-means to S as a post-processing step to obtain a community partition

Modularity Maximization: an Example



$$B = \begin{bmatrix} -0.32 & 0.79 & 0.68 & 0.57 & -0.43 & -0.43 & -0.43 & -0.32 & -0.11 \\ 0.79 & -0.14 & 0.79 & -0.29 & -0.29 & -0.29 & -0.29 & -0.21 & -0.07 \\ 0.68 & 0.79 & -0.32 & 0.57 & -0.43 & -0.43 & -0.43 & -0.32 & -0.11 \\ 0.57 & -0.29 & 0.57 & -0.57 & 0.43 & 0.43 & -0.57 & -0.43 & -0.14 \\ -0.43 & -0.29 & -0.43 & 0.43 & -0.57 & 0.43 & 0.43 & 0.57 & -0.14 \\ -0.43 & -0.29 & -0.43 & 0.43 & 0.43 & -0.57 & 0.43 & 0.57 & -0.14 \\ -0.43 & -0.29 & -0.43 & -0.57 & 0.43 & 0.43 & -0.57 & 0.57 & 0.86 \\ -0.32 & -0.21 & -0.32 & -0.43 & 0.57 & 0.57 & 0.57 & -0.32 & -0.11 \\ -0.11 & -0.07 & -0.11 & -0.14 & -0.14 & -0.14 & 0.86 & -0.11 & -0.04 \end{bmatrix}$$

Modularity Matrix

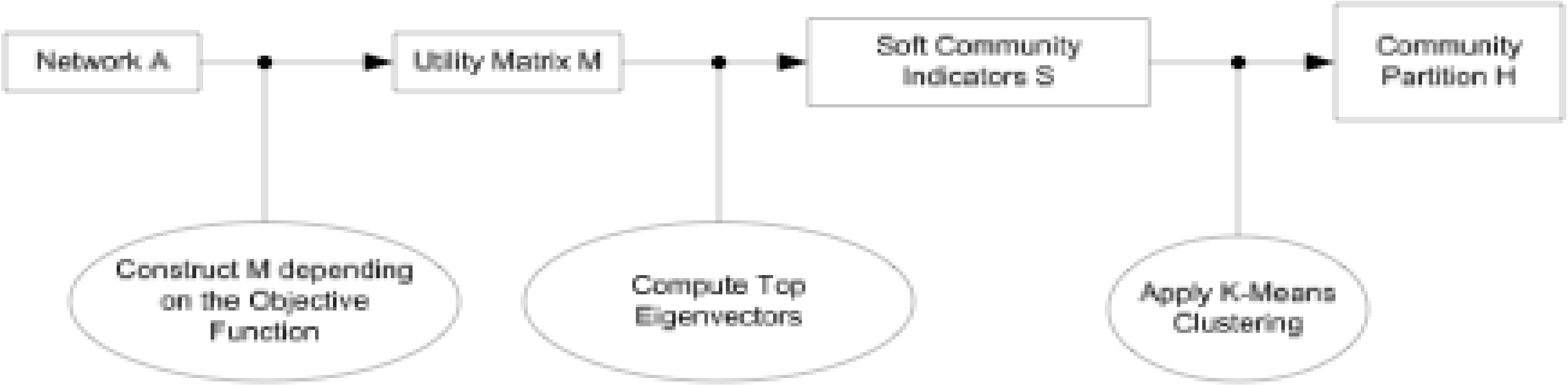
$$S = \begin{bmatrix} 0.44 & -0.00 \\ 0.38 & 0.23 \\ 0.44 & -0.00 \\ 0.17 & -0.48 \\ -0.29 & -0.32 \\ -0.29 & -0.32 \\ -0.38 & 0.34 \\ -0.34 & -0.08 \\ -0.14 & 0.63 \end{bmatrix}$$

k-means

Two Communities:
{1, 2, 3, 4} and {5, 6, 7, 8, 9}

A Unified View for Community Partition

Latent space models, block models, spectral clustering, and modularity maximization can be unified as



$$\text{Utility Matrix } M = \begin{cases} \text{modified proximity matrix } \widetilde{P} & \textit{if} \text{ latent space models} \\ \text{adjacency matrix } A & \textit{if} \text{ block models} \\ \text{graph Laplacian } \widetilde{L} & \textit{if} \text{ spectral clustering} \\ \text{modularity maximization } B & \textit{if} \text{ modularity maximization} \end{cases}$$

A Unified View for Community Partition: Matrix Factorization Form

The above network-centric community detection methods (latent space models, block models, spectral clustering, and modularity maximization) can be formulated as:

$$\begin{aligned} \max(\min)_S \quad & Tr(S^T X S) \\ s.t. \quad & S^T S = I \end{aligned}$$

$$X = \begin{cases} \Delta(D) & \text{(Latent Space Models)} \\ \text{Sociomatrix} & \text{(Block Model Approximation)} \\ \text{Graph Laplacian} & \text{(Cut Minimization)} \\ \text{Modularity Matrix} & \text{(Modularity maximization)} \end{cases}$$

Limitation: Requires the user to specify the number of communities beforehand

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Hierarchy-Centric Community Detection

Goal: build a hierarchical structure of communities based on network topology

Allow the analysis of a network at different resolutions

Representative approaches:

- Divisive Hierarchical Clustering – e.g., Girvan-Newman ($O(m^2 n)$)
- Agglomerative Hierarchical clustering – e.g., Louvain ($O(n \log n)$)

Divisive Hierarchical Clustering

Divisive clustering

- Partition nodes into several sets
- Each set is further divided into smaller ones
- Network-centric partition can be applied to the partition

One particular example: **recursively remove the "weakest" tie**

- Find the edge with the least strength
- Remove the edge and update the corresponding strength of each edge

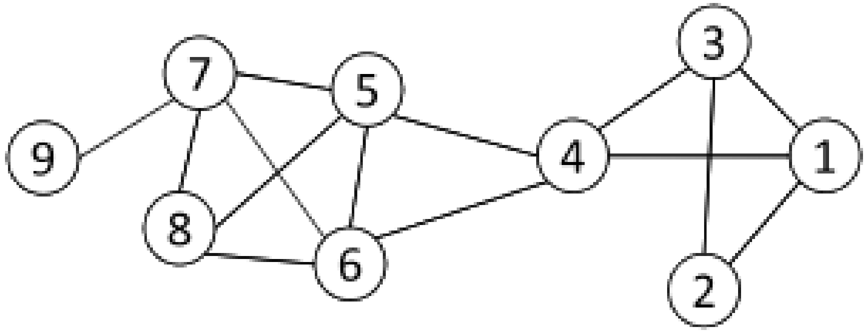
Recursively apply the above two steps until a network is decomposed into the desired number of connected components.

Each component forms a community

Edge Betweenness

The strength of a tie can be measured by edge betweenness

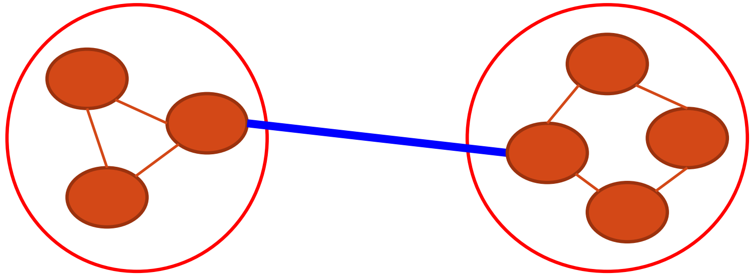
Edge betweenness: the fraction of shortest paths that pass along the edge



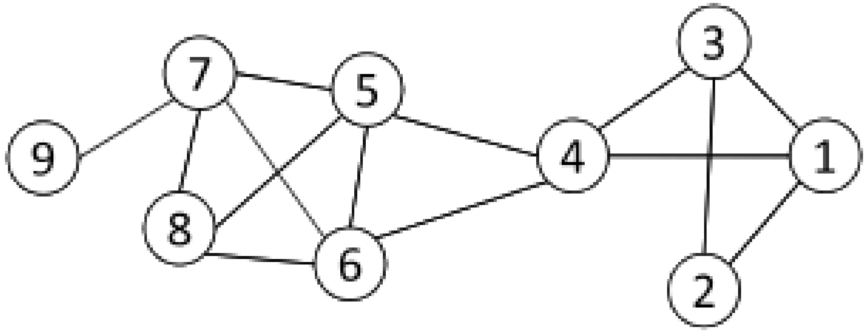
$$\text{edge-betweenness(e)} = \Sigma_{s<t} \frac{\sigma_{st}(e)}{\sigma_{s,t}}$$

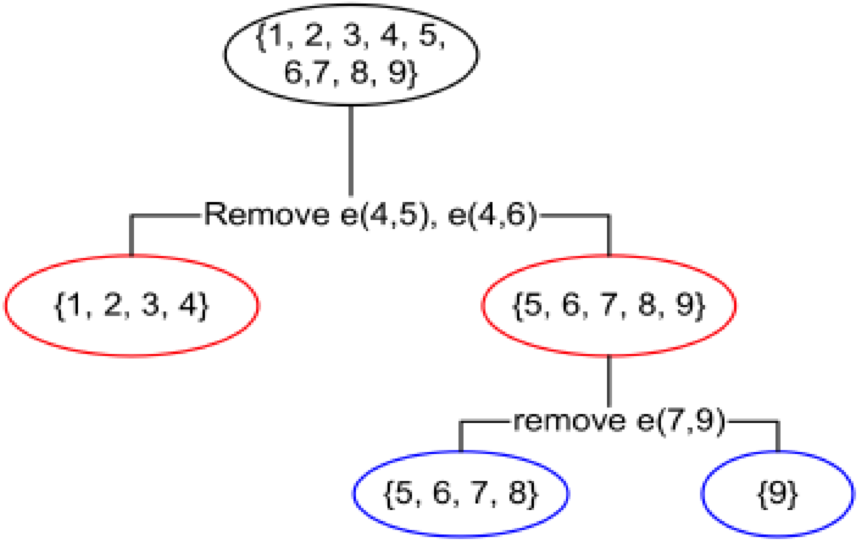
The edge betweenness of e(1, 2) is 4, as all the shortest paths from 2 to {4, 5, 6, 7, 8, 9} have to either pass e(1, 2) or e(2, 3), and e(1,2) is the shortest path between 1 and 2

The edge with higher betweenness tends to be the bridge between two communities.



Divisive Clustering Based on Edge Betweenness





Initial betweenness value

Table 3.3: Edge Betweenness

	1	2	3	4	5	6	7	8	9
1	0	4	1	9	0	0	0	0	0
2	4	0	4	0	0	0	0	0	0
3	1	4	0	9	0	0	0	0	0
4	9	0	9	0	10	10	0	0	0
5	0	0	0	10	0	1	6	3	0
6	0	0	0	10	1	0	6	3	0
7	0	0	0	0	6	6	0	2	8
8	0	0	0	0	3	3	2	0	0
9	0	0	0	0	0	0	8	0	0

After removing e(4,5), the betweenness of e(4, 6) becomes 20, which is the highest;

After removing e(4,6), the edge e(7,9) has the highest betweenness value 4 and should be removed.

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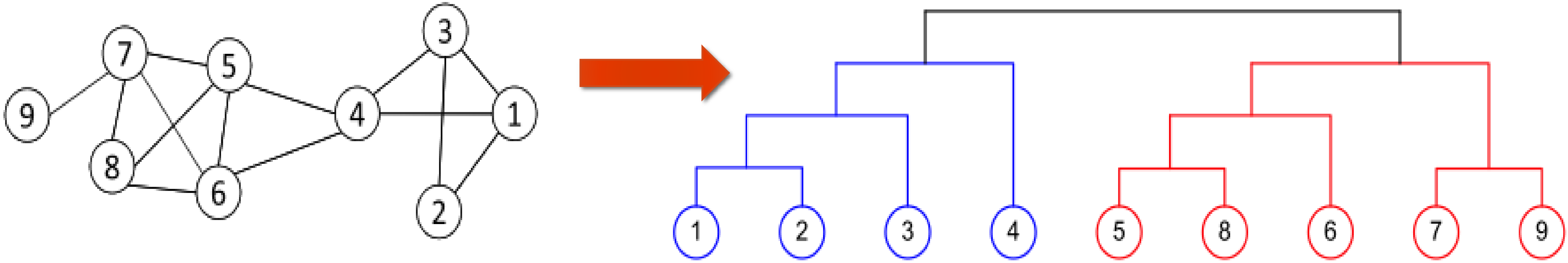
Agglomerative Hierarchical Clustering

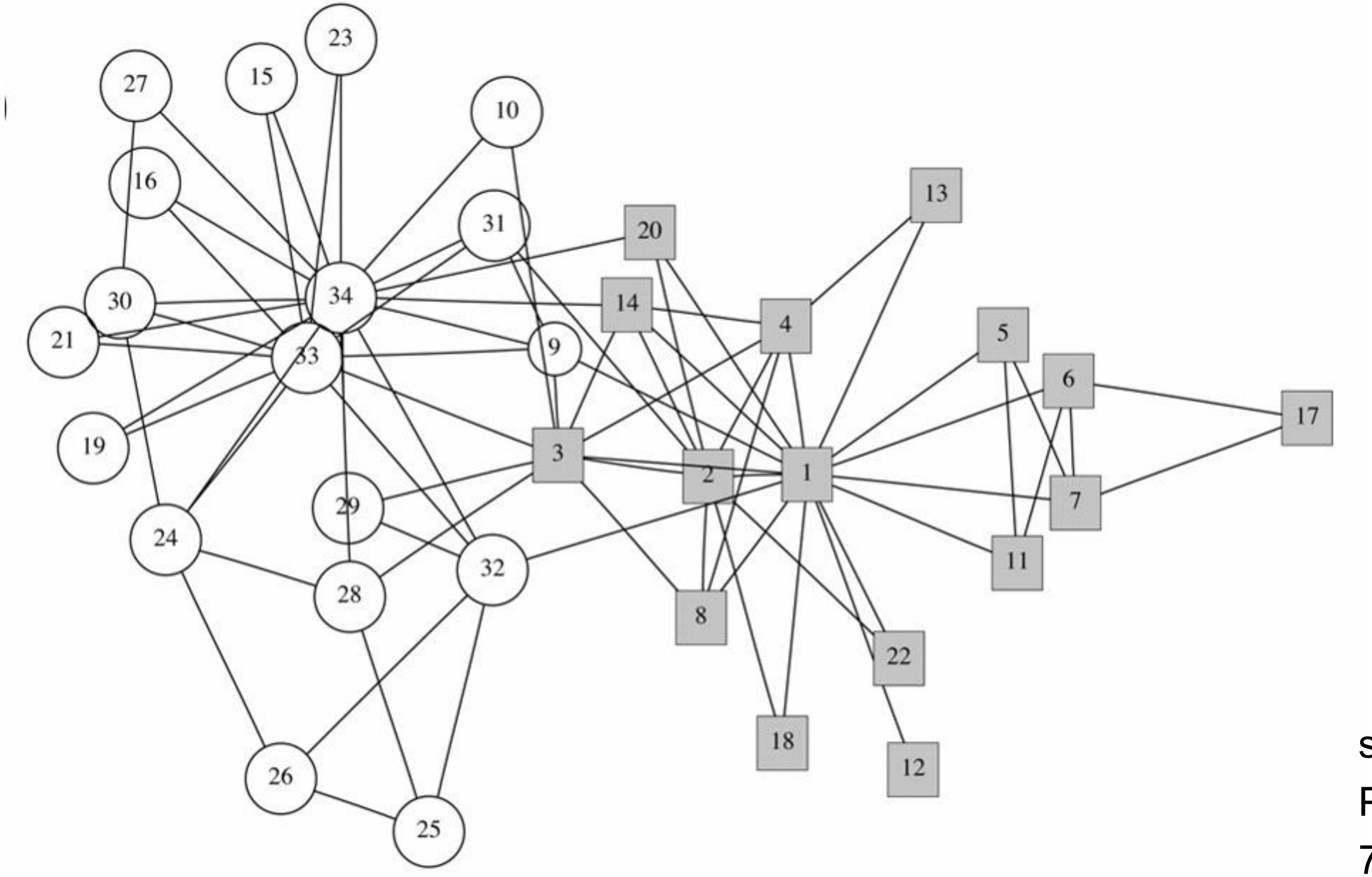
Initialize each node as a community

Merge communities successively into larger communities following a particular criterion

- e.g., based on modularity increase

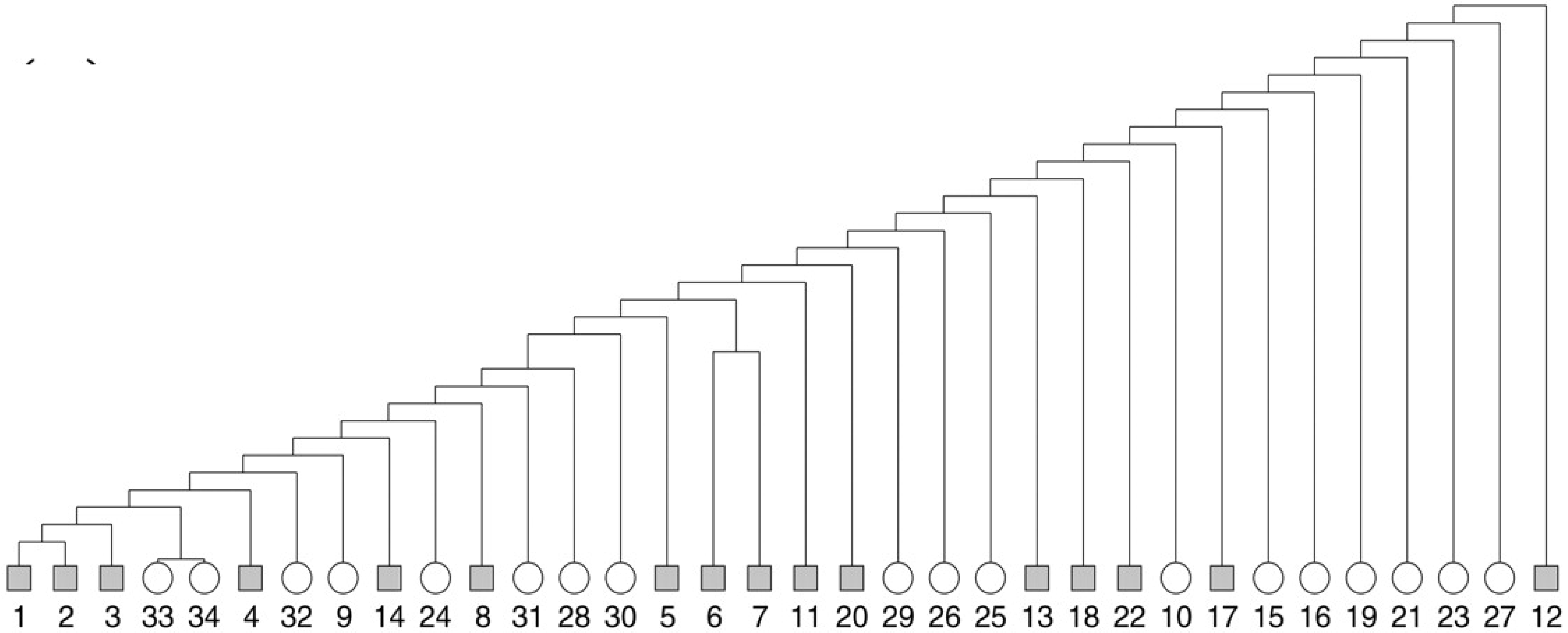


Hierarchical Clustering: Zachary Karate Club



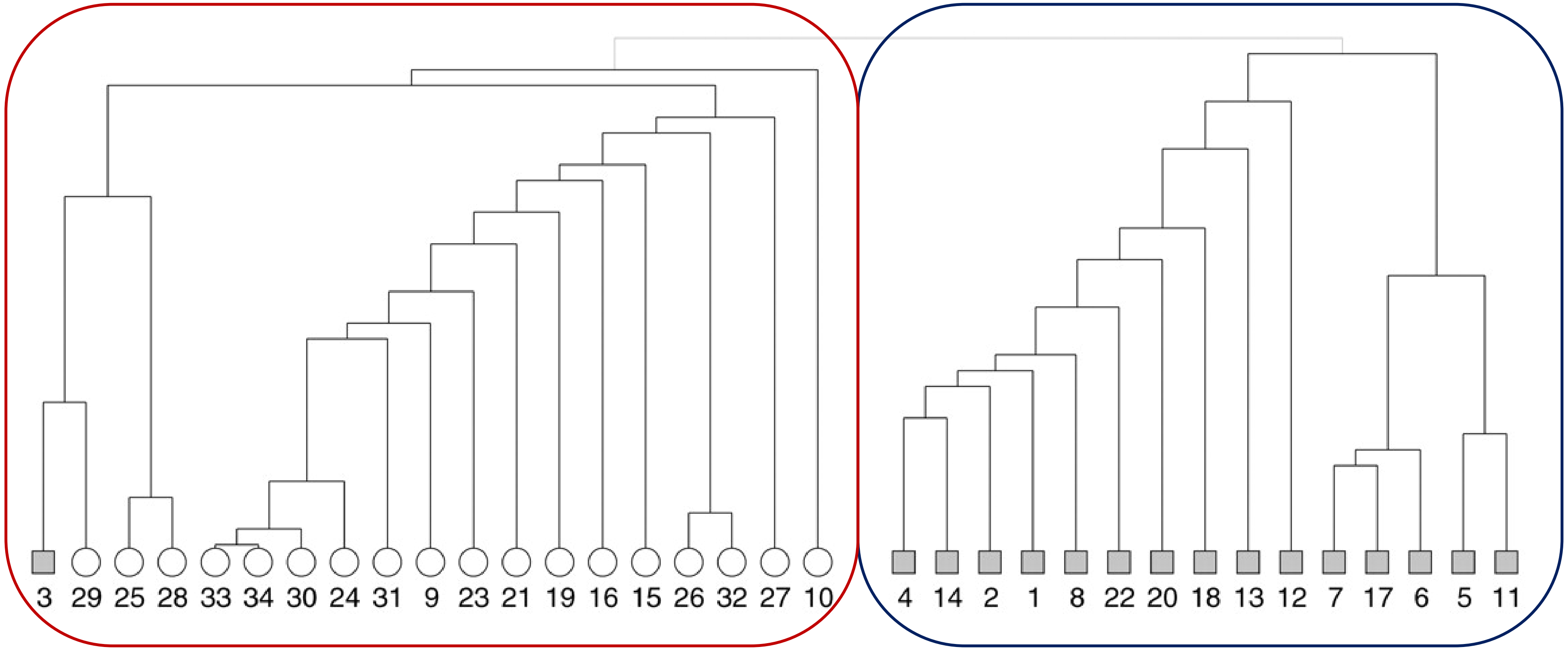
source: Girvan and Newman, PNAS June 11, 2002 99(12): 7821-7826

Is Hierarchical Clustering Really This Bad?



Zachary Karate Club data hierarchical clustering tree using edge-independent path counts

Clustering Based on Betweenness and the Karate Club Data Set



Community Detection - Summary

Node-Centric Community Detection

- *cliques, k-cliques, k-clubs*

Group-Centric Community Detection

- *quasi-cliques*

Network-Centric Community Detection

- *Clustering based on vertex similarity*
- *Latent space models, block models, spectral clustering, modularity maximization*

Hierarchy-Centric Community Detection

- *Divisive clustering*
- *Agglomerative clustering*

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Link Prediction in Social Networks

Objective:

- **predict future links between pairs of nodes in the network!**
- in commercial social networks (e.g., Facebook, LinkedIn), users are often recommended as potential friends
- both content similarity and structure may be used:
  - **Content-based measures – use the principle of homophily**
    - **Idea:** nodes with a similar content are more likely to become linked.
    - enhance link prediction, but are sensitive to the network at hand
      - e.g., short and noisy tweets with nonstandard acronyms on Twitter

    X **content-based homophily does not necessarily imply structural connectivity!**

Link Prediction in Social Networks

- **Structural measures - use the principle of triadic closure.**
  - **Idea:** ***two nodes that share their neighborhood are more likely to get connected***
  - triadic closure is very common across different domains
  - **effective in different types of networks and more directly applicable**
  - include the neighborhood-based measures, Katz measure, random walk-based measures, etc.
  - **structural connectivity usually implies content-based homophily, too**

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Neighborhood-Based Link Prediction Measures

~ use the number of common neighbors between a pair of nodes i and j, to quantify the likelihood of a future link between them.

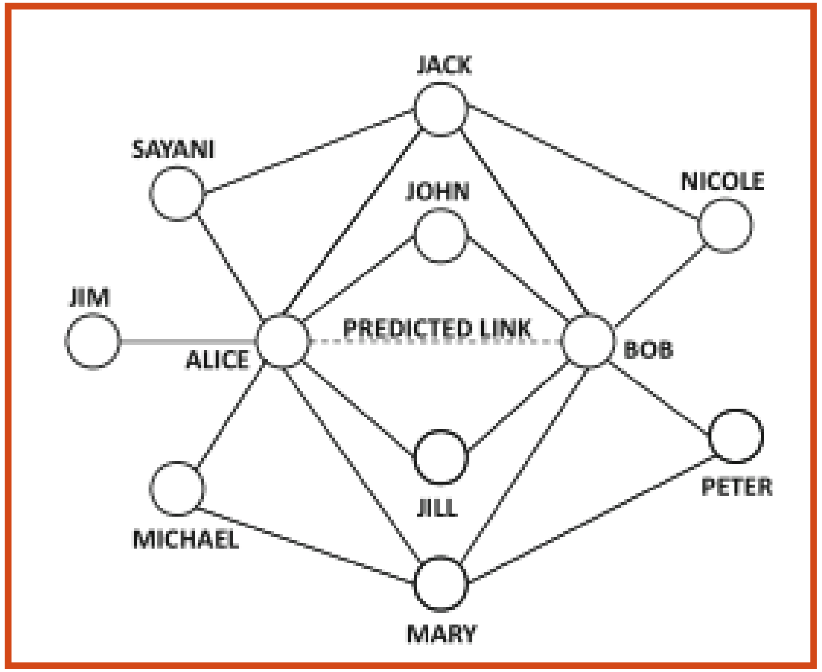
Example: Alice and Bob share 4 common neighbors.

➔ a link might eventually form between them

X Alice and Bob also have their own disjoint sets of neighbors in addition to their common neighbors

➔ account for the number and relative importance of different neighbors!

=> **normalized neighborhood-based measures**



Neighborhood-Based
Link Prediction Measures

Common Neighbor Measure between node *i* and *j* :

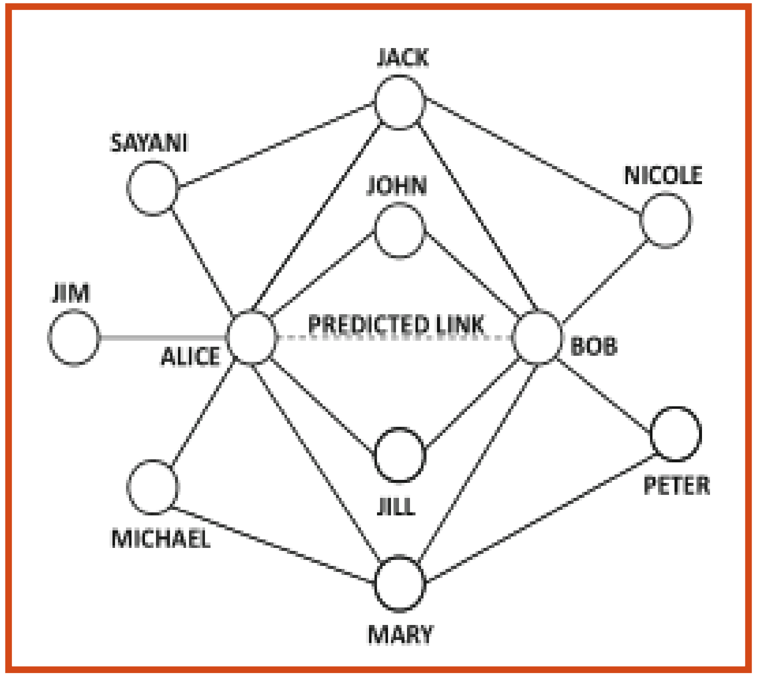
~ the number of common neighbors between nodes i and j

- **Definition:**
  Let S_i be the neighbor set of node i, and S_j the neighbor set of node j.
  The common neighbor measure between *i and j* is defined as:

$$CommonNeighbors(i,j) = |S_i \cap S_j|$$

- **Weakness:**
  - **does not account for the relative number of common neighbors as compared to the number of other connections**
  - in the example, Alice and Bob each have a relatively small node degree

  X but if Alice and Bob were very popular public figures connected to a large number of other actors, they might easily have many neighbors in common, just by chance



Neighborhood-Based Link Prediction Measures

The Jaccard-Based Link Prediction Measure between node $\boldsymbol{i}$ and $\boldsymbol{j}$:

~ measures the similarity of the neighborhood between the nodes i and j

- normalized for node degrees varying due to power-law distribution

- **Definition:**

  Let S_i be the neighbor set of node i, and S_j the neighbor set of node j.

  The Jaccard-based link prediction measure between *i and j* is defined as:

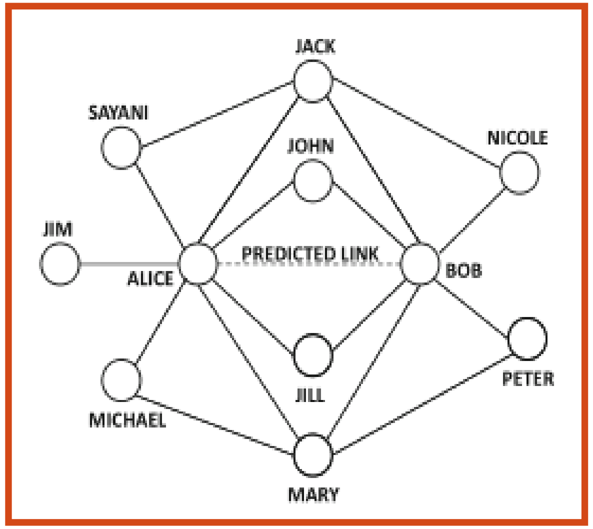
$$\boldsymbol{JaccardPredict(i,j)} = \left|\boldsymbol{S_i} \cap \boldsymbol{S_j}\right| / \left|\boldsymbol{S_i} \cup \boldsymbol{S_j}\right|$$

- **Properties:**
  - in the example, larger degrees of Alice or Bob would impact a lower Jaccard coefficient between them

  X but the Jaccard measure might not adjust well to intermediate neighbors

  - if the common neighbors of Alice and Bob (e.g., John, or Jill) would be very popular (i.e., have a high degree), then they might occur as common neighbors also of many other pairs of nodes

    => they would be less important for measuring link prediction



The Jaccard measure between Alice and Bob is $4/9$.

Neighborhood-Based Link Prediction Measures

The Adamic–Adar measure Measure between node i and j:

~ a weighted version of the common-neighbor measure

- designed to account for the varying importance of different common neighbors
- the weight of a common neighbor is a decreasing function of its node degree
  - e.g., the weight of node k can be expressed as $1/\log(|S_k|)$, where S_k is the neighbor set of node k

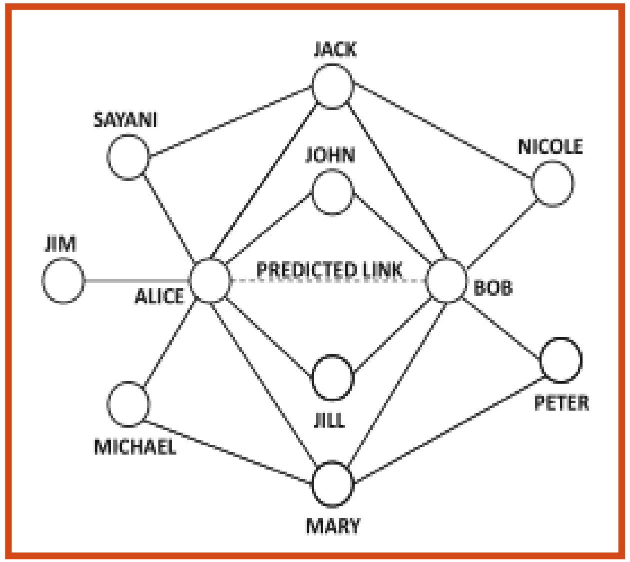
- **Definition:**

  Let S_i, S_j, and S_k, be the neighbor set of node i, j, and k, resp. **The Adamic–Adar measure between *i and j*** is equal to the weighted sum of common neighbors between the nodes i and j:

$$\boldsymbol{AdamicAdar(i,j) = \sum_k [1/\log([S_k])];\ k \in S_i \cap S_j}$$

  - in the example, the Adamic-Adar measure between Alice and Bob is:

$$1/\log(4) + 1/\log(2) + 1/\log(2) + 1/\log(4) = 3/\log(2)$$



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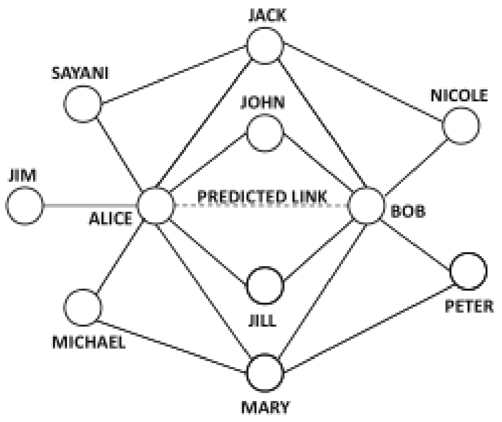
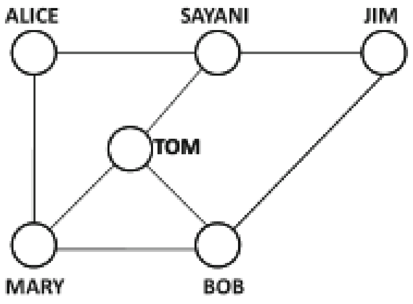


(a) Many common neighbors between Alice and Bob



(b) Many indirect connections between Alice and Bob

Link Prediction: Katz Measure

Neighborhood-based measures:

- estimate the likelihood of a link between a pair of nodes

X but not effective when the number of neighbors shared between a pair of nodes is small

Example (b):

- Alice and Bob share one neighbor in common
- Alice and Jim also share one neighbor in common

X **neighborhood-based measures have difficulty in distinguishing between these cases**

- **but there is indirect connectivity through longer paths**

➔ in such cases, walk-based measures are more appropriate for link-prediction, excellent results provides, e.g., **the Katz measure**

Link Prediction: Katz Measure

- **Definition:**

  Let $n_{ij}^{(t)}$ be the number of walks of length t between the nodes i and j. For a pre-set parameter $\beta < 1$,

  the Katz measure between i and j is defined as: $\boldsymbol{Katz(i,j) = \sum_{t=1}^{\infty} \beta^t \cdot n_{ij}^{(t)}}$

  β is a discount factor that de-emphasizes the walks of a longer length.

- for small values of β, the infinite summation will converge.

- **If A is the symmetric adjacency matrix of an undirected network, then the $n \times n$ pairwise Katz coefficient matrix K can be computed as:**

$$\boldsymbol{K = \sum_{i=1}^{\infty} (\beta A)^i = (I - \beta A)^{-1} - I}$$

  The eigenvalues of A^k are the k-th powers of the eigenvalues of A.

- **highly central nodes have the propensity to form links with many nodes**

  => The sum of the Katz coefficients of node *i* with respect to other nodes is referred to as its **Katz centrality**

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Link Prediction as a Classification Problem

the presence or absence of a link between a pair of nodes is treated as a binary class indicator

➔ **<u>construct a multi-dimensional data record for each pair of nodes</u>**

- **the features/attributes of the data record include all the different neighborhood-based, Katz-based, or walk-based similarities between nodes**
- **in addition, a number of other preferential-attachment features, such as node-degrees of each node in the pair, are used**
- **the result is a positive-unlabeled classification problem, where node pairs with edges are the positive examples, and the remaining pairs are unlabeled.**
- **unlabeled examples can be approximately treated as negative examples during training**
- **as there are too many negative example pairs in large and sparse networks, only a sample of the negative examples is used**

Link Prediction as a Classification Problem

The supervised link prediction algorithm works as follows:

1. Training phase:

- Generate the data with one data record for each pair of nodes with an edge between them, and a sample of data records from pairs of nodes without edges between them
- The features correspond to the extracted similarity and structural features between node pairs
- The class label is the presence or absence of an edge between the corresponding pair of nodes
- Construct a classifier model on the data and train it

2. Testing phase:

- Convert each test node pair to a multi-dimensional record
- Use the pre-trained classifier to make label predictions
  - **logistic regression represents a common choice for the base classifier**
  - **cost-sensitive versions of various classifiers are often used because of the imbalanced data**

Link Prediction as a Classification Problem

Advantages of the supervised approach to link prediction:

- content features can be extracted and used easily
- the classifier automatically learns the relevance of the features
- for directed networks, features can be extracted asymmetrically, too
  - e.g., indegrees and outdegrees (instead of plain node degrees in undirected networks), or various random walk-based measures (e.g., personalized PageRank)
- a higher flexibility due to the ability to learn relationships between links and features of various types

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Link Prediction - Summary

Different measures vary in their effectiveness over different data sets:

- neighborhood-based measures can be computed efficiently for extensive data sets and perform almost as well as the other unsupervised measures
- random walk-based and Katz-based measures are helpful for very sparse networks, in which the number of common neighbors cannot be robustly measured
- supervision provides better accuracy and adaptability across various domains of social networks, but is computationally expensive
- in recent years, content has also been used to enhance link prediction, yet structural measures are far more powerful (due to triadic properties of real networks).
- content-based measures are based on “reverse homophily,” where similar or link-correlated content is leveraged for predicting links

Thoughts on Design

SNA theory has inspired some of the first SNSs (e.g., SixDegrees), but it is still not used so often in conjunction with design decisions such as:

- **How can an online social media platform (or its administrators) leverage the methods and insights of SNA?**
- **How can a platform encourage its users to act (locally) in such a way that would expand the community?**
- **What measures can an online community take to optimize its structure?**

  Example: cliques can be undesirable because they shun newcomers

- **What are the desirable structures for different types of online platforms?**
- **How can online communities identify and utilize key players for their benefit?**

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- Homophily, Social Influence, and Affiliation
- Link Analysis
- **Real-World Complex Networks, Community Detection, and Link Prediction**